

HANYANG UNIVERSITY · DEPT. OF NUCLEAR ENGINEERING · THINK LAB · NUE4067

# Finding SMR Data, and PWR-type SMRs

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From ARIS, NRC and NEA to AI - then reading **NuScale** and **i-SMR** side by side

Weeks 3-4 · Instructor **Minseop Song** · Rev1.0 · September 2026

# Values that disagree - three real examples

- The instruction every new SMR project gives in week one - “Put together the NuScale and i-SMR specs”
- Search and the material pours in, but **the numbers disagree**. With no basis to judge, you end up copying the most recent article
- Then the question comes - “Where did that number come from?” No answer, and **the whole table is worthless**

i-SMR thermal power per module

**520 MWth**

some domestic sources

**540 MW(th)**

IAEA ARIS design report

NuScale electrical output per module

**50 MWe**

2020, US600

**60 MWe**

interim revision

**77 MWe**

2025, US460

NuScale primary operating pressure

**12.8 MPa**

US600

**13.8 MPa**

US460 = 2 000 psia

None of the three is a “bad source” - they count different things, at different times

# How this deck is built - Part I data, Part II designs

- **Part I** is the procedure for pinning a number down; **Part II** applies it to three PWR-type SMRs
- The comparison table in ch. 8 is **Part I filled in cell by cell** - what could not be confirmed stays blank

## Part I · Pin the data down

- 1 Where the data lives - three tiers of sources
- 2 The three legacy sources - ARIS · NRC · NEA
- 3 Collecting with AI - how far to let it go
- 4 Cross-checking - how AI gets numbers wrong

## Part II · Read the designs

- 5 PWR, BWR, and going integral
- 6 NuScale VOYGR - the pump taken out
- 7 i-SMR - the pump kept in
- 8 Head to head - the two integral designs
- 9 Control case BWRX-300 - boil to circulate
- 10 Wrap-up - one problem, three answers

# 01 Where the data lives

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No calculations here. Instead we set the yardstick for judging sources that every later chapter will use. Once you know why the same quantity gets written differently, conflicting data stops being alarming.

# Four reasons the same value is written differently

- In all four cases **neither side is false** - they count different things at different times
- So the first question is not "is it right?" but "**what did it count?**"

## ① The design changes

NuScale went from 160 to 250 MWt per module. 50 MWe in a 2020 source and 77 MWe in a 2025 one are both right - the first is US600, the second US460.

## ② They count different things

Thermal or electrical? Per module or per plant? i-SMR's 170 MWe is per module, 680 MWe is four of them. Mix the two and you are off by a factor of four.

## ③ They were written at different times

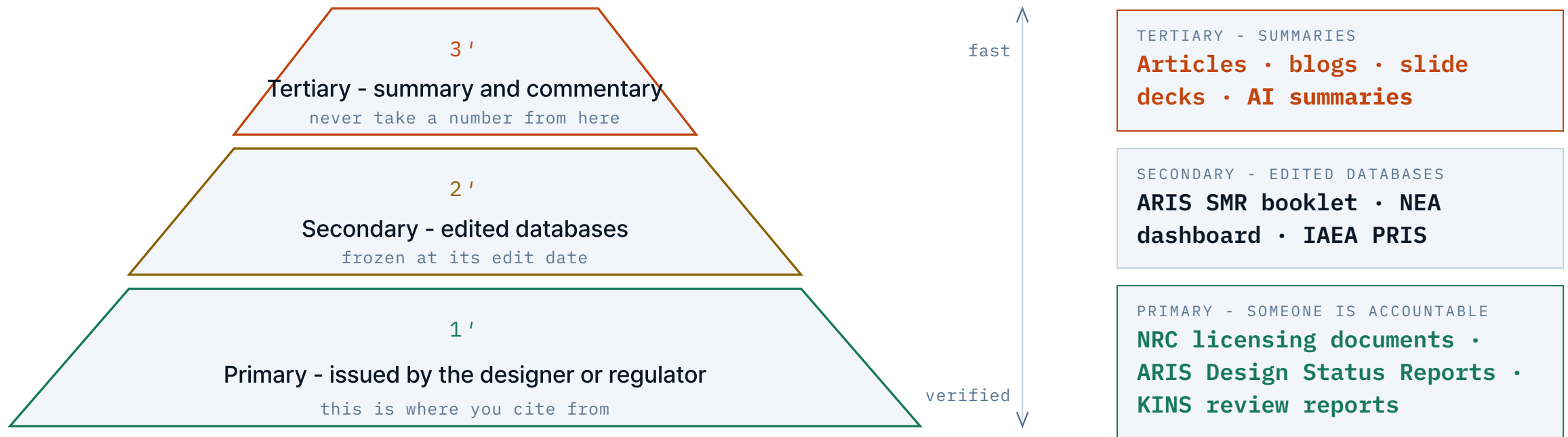
The IAEA SMR booklet is refreshed each edition, but if the latest is 2024 then nothing from 2026 is in it.

## ④ A copy of a copy

Article A makes a typo and B through Z carry it. Twenty search hits can trace to one origin. A number appearing in many places is not evidence that it is right.

# Three tiers of sources - primary, secondary, tertiary

- The higher you go the **faster** it is; the lower, the more **verified**. Explore at the top, cite from the bottom
- Anything an AI summarises is **tertiary** - fine for orientation, never for numbers

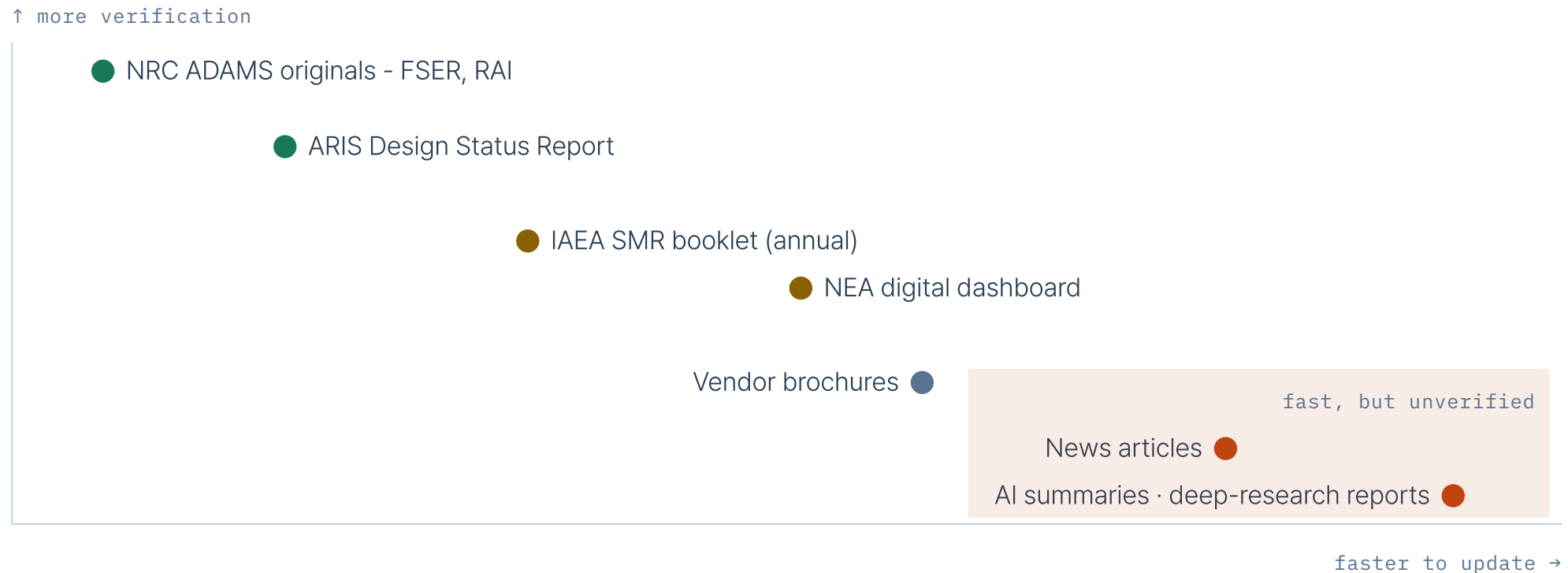


ARIS Advanced Reactors Information System • DSR Design Status Report • PRIS Power Reactor Information System • KINS Korea Institute of Nuclear Safety

Source This textbook §1.2, Fig. 1.1 • a number seen in a tertiary source must be traced down one tier before it is cited

# Update speed against verification depth

- AI collection is attractive and dangerous for the same reason - it sits in **the fastest, least verified corner**
- So the working rule is simple: **explore on the fast side, cite from the slow side**



# The minimum record for a value - five fields

- Write value, unit, source, access date and note **on one line** and you never lose a day to “where did this come from?”
- The **access date** matters most - for fast-ageing values like licensing status it is as important as the value

Value

540

Unit · definition

MW(th), rated core thermal power per module

Source

IAEA ARIS, i-SMR Design Status Report

Access date

2026-09-16

Note

Some domestic sources say 520 MWth - see ch. 4

If the procedure feels tedious, consider **why safety analysis reports are so thick**. Much of a licensing document exists to prove where each number came from. Establishing **the provenance and uncertainty of the inputs** takes longer than the calculation itself.

Nuclear link · this habit is the doorway to V&V

# 02

## The three legacy sources

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ARIS, the US NRC, and the OECD/NEA dashboard. Lump them together as “SMR databases” and you will search all three every time. Know which question each one answers and you open only one.

# What each of the **three sources** answers

- Treat them as one and you search all three every time. **Question** → **source** is a fixed mapping
- Knowing what each **cannot** do matters more - looking there wastes the afternoon

## IAEA ARIS

answers

What are this design's specifications?

cannot do

How far this project has got today

## US NRC

answers

What has the regulator reviewed and approved?

cannot do

Designs outside the United States

## OECD/NEA

answers

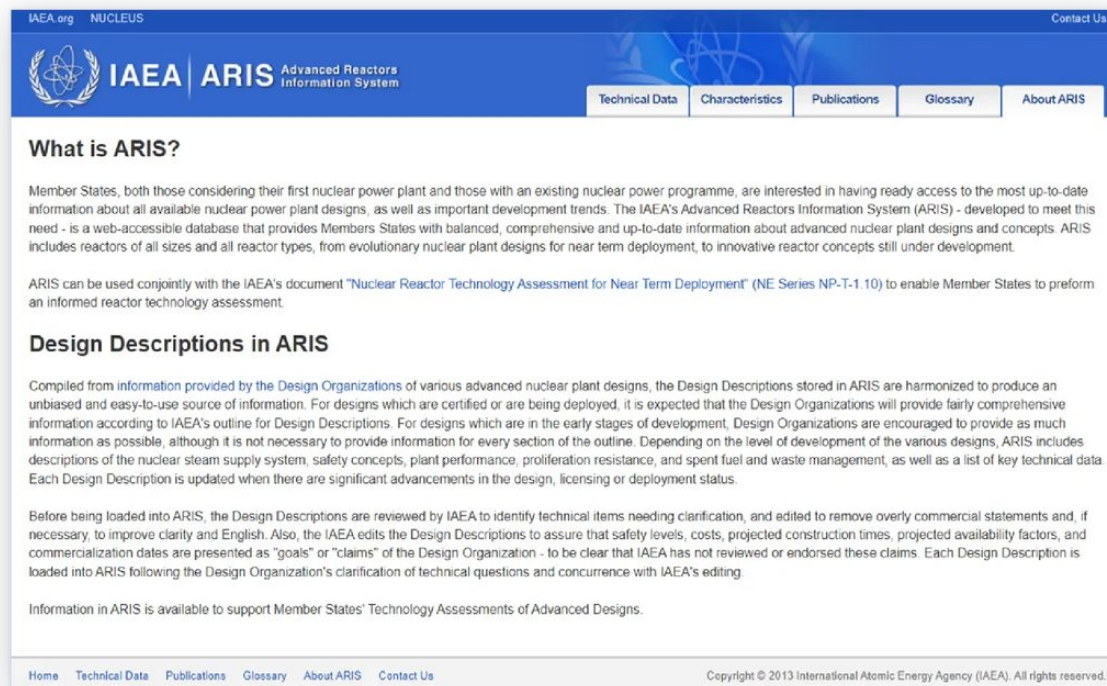
How close is this design to deployment?

cannot do

The technical detail of the design

# ARIS - the Design Status Report

- You can download the Design Status Report **written and submitted by the designer** - someone is accountable for the numbers
- It follows the IAEA template, so **pressure, temperature, flow and fuel** are all there. Even for Korean designs, **ARIS beats the vendor brochure**



The ARIS landing page - the "Design Descriptions in ARIS" note states the **nature and limits** of what designers file

**DSR** Design Status Report - written and filed by the designer itself · **HTGR** High-Temperature Gas-cooled Reactor · **SFR** Sodium-cooled Fast Reactor · **MSR** Molten Salt Reactor

**Source** This textbook §2.2, Fig. 2.1 · the IAEA ARIS landing page at <https://aris.iaea.org> · accessed September 2026

## SEARCH FILTERS

Type (PWR, BWR, HTGR, SFR, MSR)  
 • coolant • moderator • design status • country • power range

## WHAT YOU GET

The Design Status Report as a PDF, plus the SMR booklet

A brochure is **promotional** and drops fields; a press release carries no specifications at all. That is why the i-SMR table in chapter 7 was built from the ARIS report.

# ARIS search results - design status by country

- Each row carries design, organisation, coolant, moderator, **design status**, country and purpose. The column to read is **design status**
- As of 2026 the SMRs **actually making electricity** are in Russia and China. Korea's entries sit in the design phase

Advanced Reactors Information System(ARIS)

OverviewGeneral dataNuclear Steam Supply SystemReactor Coolant SystemReactor CoreCore MaterialsReactor Pressure Vessel

Type

☐ All☐ PWR☐ BWR☐ HWR☐ SCWR☐ IPWR☐ GCR☐ GFR☐ SFR☐ LFR☐ MSR☐ FR☒ SMR

Country

☐ All☐ Canada☐ China☐ EU☐ France☐ India☐ Japan☐ Rep. of Korea☒ Russia☐ USA☐ Other

Status

☒ All☐ On Hold☐ Under Design☐ Licensed☐ Construction☐ In Operation

Purpose

☒ All☐ Commercial☐ Demonstration☐ Experimental☐ Prototype

(Click on acronym for more information)

| OVERVIEW         |                                            |                  |                             |              |               |         |      |               |
|------------------|--------------------------------------------|------------------|-----------------------------|--------------|---------------|---------|------|---------------|
| Acronym          | Full name                                  | Design Org.      | Coolant                     | Moderator    | Design Status | Country | Type | Purpose       |
| KLT-40S          | KLT-40S                                    | OKBM             | Light Water                 | Light Water  | Construction  | Russia  | PWR  | Commerical    |
| VBER-300         | VBER-300                                   | OKBM             | Light Water                 | Light Water  | Under Design  | Russia  | PWR  | Commerical    |
| VVER-600 (V-498) | VVER-600 (V-498)                           | Gidropress       | Light Water                 | Light Water  | Under Design  | Russia  | PWR  | Commerical    |
| VVER-640 (V-407) | VVER-640 (V-407)                           | Gidropress       | Light Water                 | Light Water  | Under Design  | Russia  | PWR  | Commerical    |
| MBIR             | Multipurpose fast-neutron research reactor | NIKIET           | Sodium                      | No Moderator | Under Design  | Russia  | SFR  | Experimental  |
| BREST-OD-300     | BREST-OD-300                               | RDIPE            | Lead                        | No Moderator | Under Design  | Russia  | LFR  | Demonstration |
| SVBR-100         | SVBR-100                                   | AKME Engineering | Lead Bismuth Eutectic alloy | No Moderator | Under Design  | Russia  | LFR  | Commerical    |

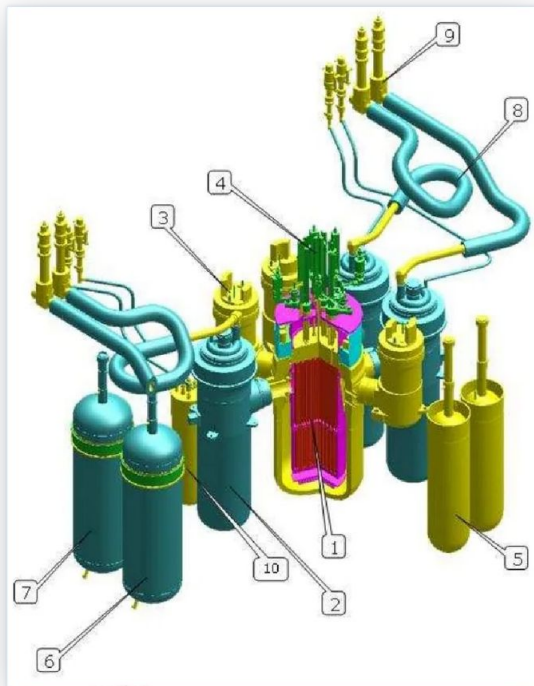
Russia - the design-status column shows **Operating** and **Under Construction**

| OVERVIEW      |                                                                 |                                          |                             |              |                         |         |      |               |
|---------------|-----------------------------------------------------------------|------------------------------------------|-----------------------------|--------------|-------------------------|---------|------|---------------|
| Acronym       | Full name                                                       | Design Org.                              | Coolant                     | Moderator    | Design Status           | Country | Type | Purpose       |
| NuScale       | NuScale SMR                                                     | NuScale Power Inc.                       | Light Water                 | Light Water  | Under Regulatory Review | USA     | IPWR | Commerical    |
| Prismatic HTR | Prismatic Modular High Temperature GCR                          | General Atomics                          | Helium                      | Graphite     | Under Design            | USA     | GCR  | Commerical    |
| PRISM         | Power Reactor Innovative Small Reactor                          | GE-Hitachi                               | Sodium                      | No Moderator | Under Design            | USA     | SFR  | Commerical    |
| TWR-P         | Travelling Wave Reactor-Prototype                               | TerraPower                               | Sodium                      | No Moderator | Under Design            | USA     | SFR  | Commerical    |
| G4M           | Gen4 Module                                                     | Gen4 Energy Inc.                         | Lead Bismuth Eutectic alloy | No Moderator | Under Design            | USA     | LFR  | Commerical    |
| Mk1 PB-FHR    | Mark 1 Pebble-Bed Fluoride-Salt-Cooled High Temperature Reactor | University of California, Berkeley       | Fluoride Salts              | Graphite     | Under Design            | USA     | MSR  | Commerical    |
| SmAHTR        | Small fluoride salt-cooled High Temperature Reactor             | Oak Ridge National Laboratory            | Fluoride Salts              | Graphite     | Under Design            | USA     | MSR  | Demonstration |
| ThorCon       | ThorCon                                                         | ThorCon US, Inc.                         | Molten Salt                 | Graphite     | Detailed Design         | USA     | MSR  | Commercial    |
| LFTR          | Liquid Fluoride Thorium Reactor                                 | Flibe Energy                             | Fluoride Salts              | Graphite     | Conceptual Design       | USA     | MSR  | Commercial    |
| BWRX-300      | Boiling Water Reactor X-300                                     | GE-Hitachi and Hitachi GE Nuclear Energy | Light Water                 | Light Water  | Conceptual Design       | USA     | BWR  | Commercial    |
| EM2           | Energy Multiplier Module                                        | General Atomics                          | Helium                      | No Moderator | Conceptual Design       | USA     | GFR  | Commercial    |
| SC-HTGR       | Steam Cycle High Temperature Gas-cooled Reactor                 | Framatome                                | Helium                      | Graphite     | Conceptual Design       | USA     | GCR  | Commercial    |
| W-LFR         | Westinghouse Lead-cooled Fast Reactor                           | Westinghouse Electric Company LLC        | Lead                        | No Moderator | Conceptual Design       | USA     | LFR  | Demonstration |

United States - the most concepts, but nearly all **Conceptual Design**

# Floating plants - the KLT-40S and Akademik Lomonosov

- Putting a reactor on a ship came out of **icebreaker experience**. It was designed to desalinate as well as generate
- A land design cannot simply be moved offshore - **this is why the KLT-40S uses forced circulation**. It bears directly on THINK Lab's floating-plant work



The KLT-40S primary layout - reactor, steam generator and pump tied together by **short pipes**



Akademik Lomonosov - a floating plant carrying two KLT-40S units

**KLT-40S** a small Russian PWR descended from icebreaker reactors · **driving head** the pressure difference that sustains natural circulation · **motion** the attitude changes of a vessel at sea

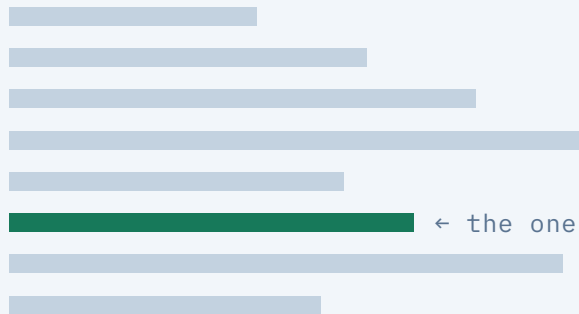
**Source** This textbook §2.2.1, Figs. 2.5-2.6 · two KLT-40S units, about 35 MWe each (70 MWe together), commercial operation May 2020

# US NRC - finding documents in ADAMS

- Nearly every document the NRC exchanged is public - applications, RAIs, responses, the FSER. **The deepest public material there is**
- Keywords return hundreds of hits and not the one you want. The trick is **the reference list of one FSER**, which carries every accession number that matters

the usual way

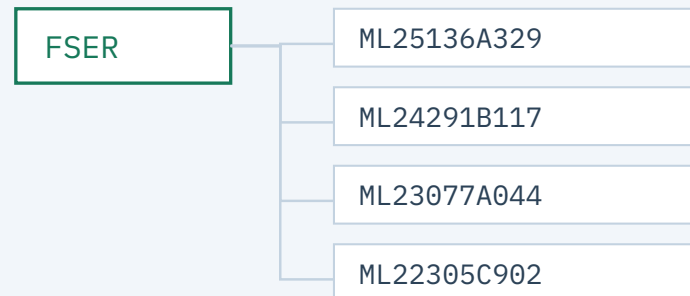
## Keyword search



Hundreds of hits come back

the faster way

## Document to document



The accession numbers are all listed there



The NRC SMR page - light-water designs, ongoing licensing, and rulemaking

What matters is not the web page but **ADAMS behind it**: the application, the NRC's questions and the answers, and the final report.

**ADAMS** Agencywide Documents Access and Management System · **RAI** Request for Additional Information · **FSER** Final Safety Evaluation Report · **ML**

**number** ADAMS accession number

**Source** This textbook §2.3, Fig. 2.7 · **open by accession number** <https://www.nrc.gov/docs/MLxxxx/MLxxxxxxxxx.pdf>

# The NEA SMR dashboard - print edition to digital

- The printed edition is **frozen at its cut-off date**; the digital one opens submission windows through the year (14 vendors in the latest round)
- **Absent from the dashboard ≠ no such design** - a vendor can ask to be left out. 79 of 129 tracked designs are listed

Printed edition (annual)



Digital dashboard (rolling)



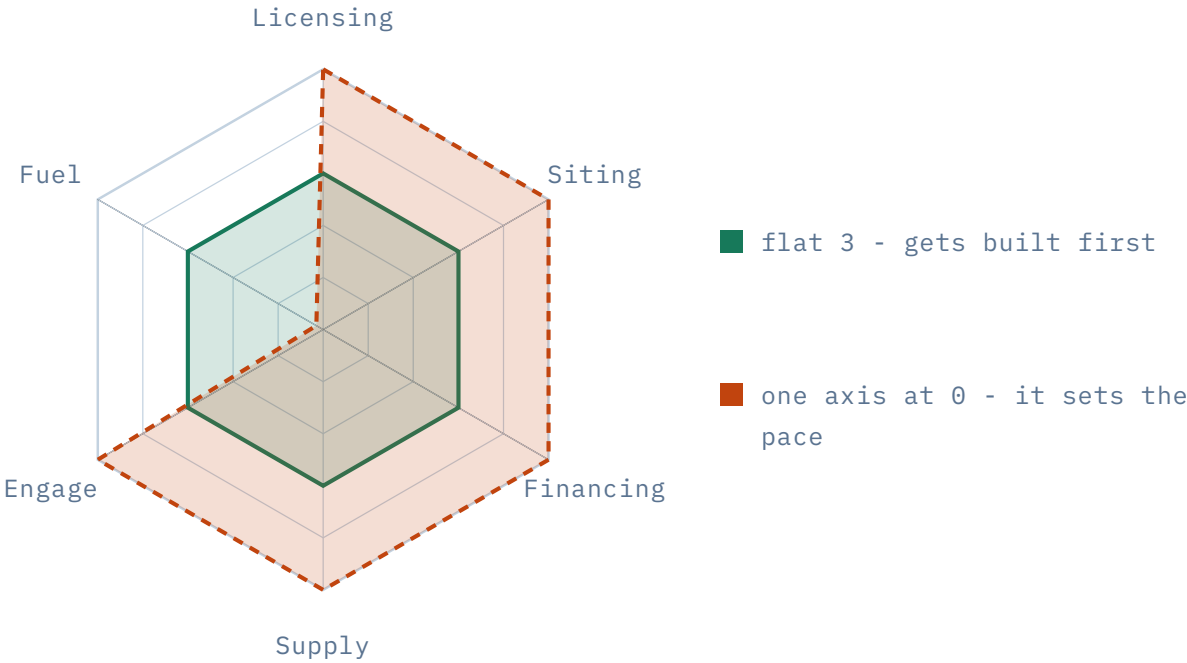
79 of the 129 designs NEA tracks are listed on the dashboard

listed 79

excluded 50 - opted out or dormant

# The dashboard's six assessment dimensions

- Between a design scoring a flat 3 on all six and one scoring 5 on five axes and 0 on one, **the flat one gets built first**
- A finished design with no fuel, or a licensed design with no money, is **the common case**



|              |                                                       |
|--------------|-------------------------------------------------------|
| Licensing    | Pre-application, design approval, construction permit |
| Siting       | Site owner's decision, permitting readiness           |
| Financing    | Nature and size of commitments, contracts, investment |
| Supply chain | MOU → binding contract → joint venture                |
| Engagement   | Community agreements, hearings, benefit sharing       |
| Fuel         | Availability; if HALEU is needed, how it is secured   |

HALEU High-Assay Low-Enriched Uranium, 5-20% enrichment · MOU non-binding memorandum of understanding · engagement agreements with the community and stakeholders

Source This textbook §§2.4.2-2.4.3, Table 2.2 · what is scored is deployment maturity, not technical maturity

# What a design card shows, and how to read it

- One card per design. The hexagon's axes are **the six dimensions**, the gauges beside it are the sub-scores, and below sit thermal power, outlet temperature, spectrum and fuel form



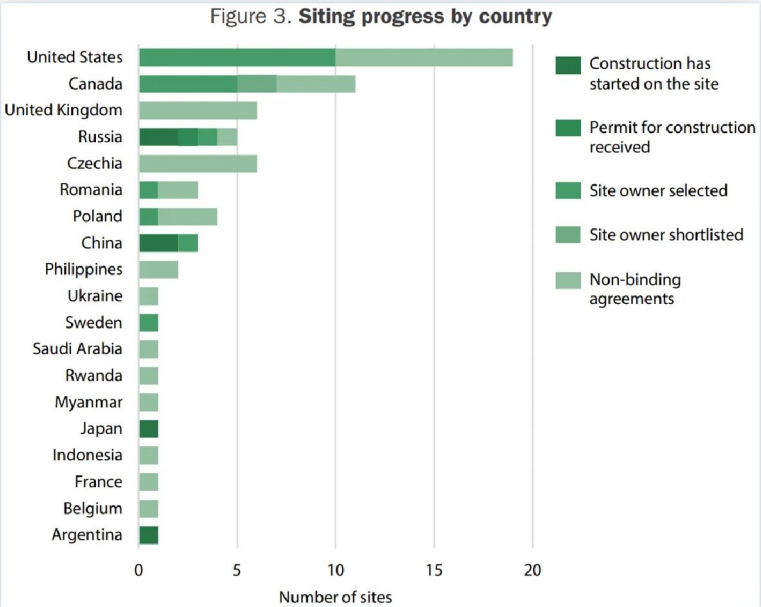
Design card - NuScale VOYGR



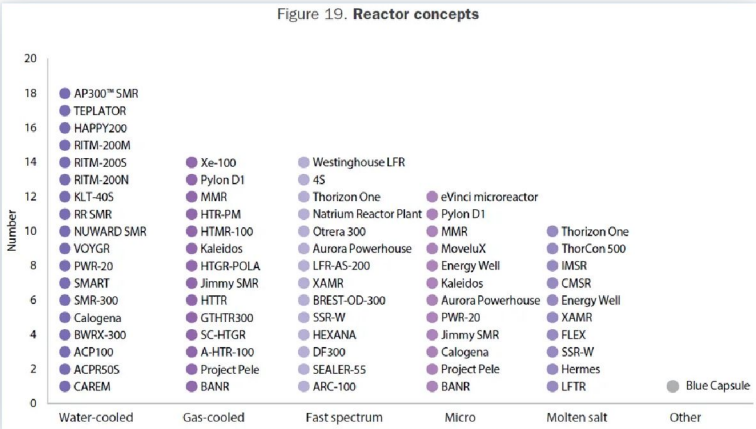
Design card - BWRX-300

# The 2024 tally from the printed second edition

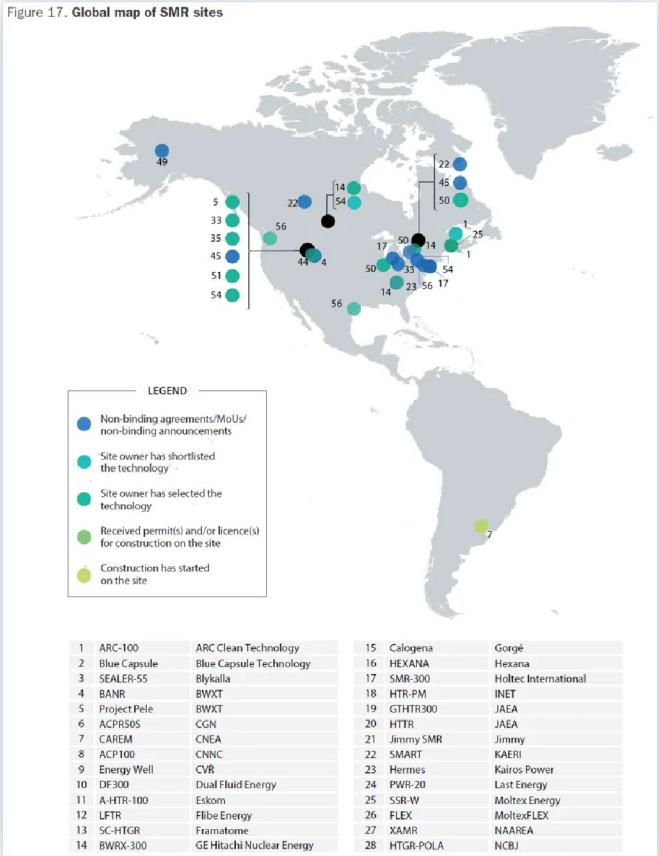
- Do not cite these directly - they are the 2024 tally. They are shown because of the **structure** they reveal
- Russia and China **lead on first-of-a-kind deployment**, with North America and Europe catching up fast on siting



Siting progress by country



Concepts by type

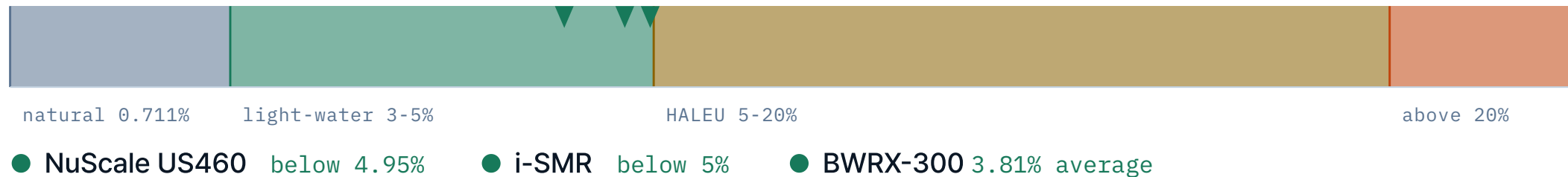


The global site map

# Fuel - enrichment requirements and the HALEU bottleneck

- Gas-cooled, fast and molten-salt designs mostly need **HALEU**. It is a proven fuel, but until 2023 there was **no commercial supply** in OECD countries
- All three light-water SMRs sit **below 5%** and use the existing fuel supply chain. That is the practical reason they go first

## The uranium enrichment ladder



### Designs in the 3-5% band

They use the existing fuel supply chain. Fuel never stops them.

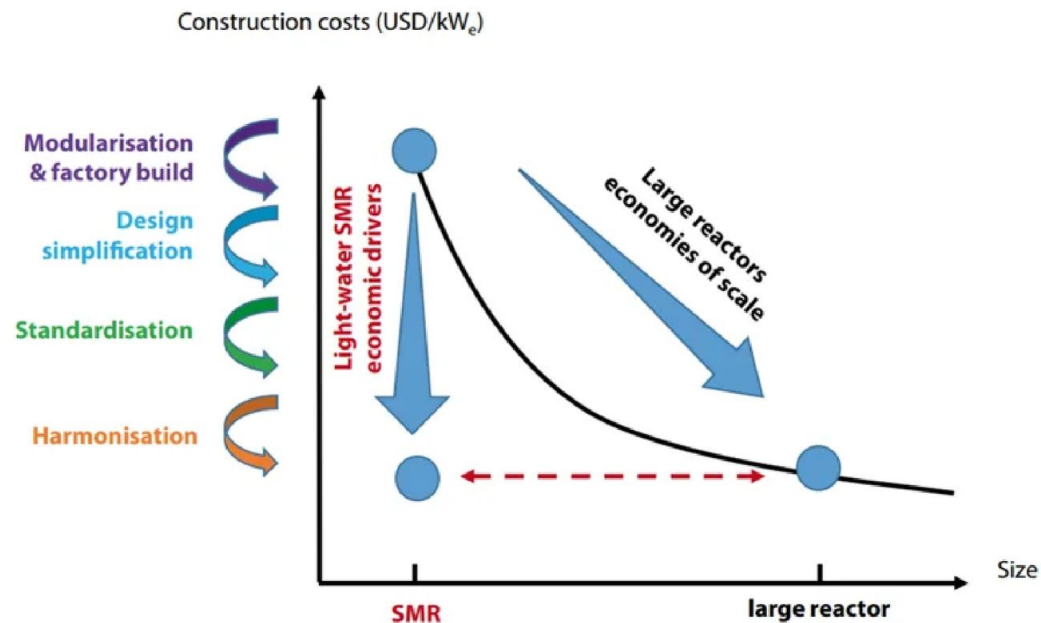
### Designs that need HALEU

Even with the design finished, a fuel route must be built first. No OECD supply until 2023.

# SMR economics - the four offsetting drivers

- Because  $n$  is below 1, cutting output raises the cost per kW - the **diseconomy of scale**
- Winning it back through **modularisation, simplification, standardisation and harmonisation** is the whole SMR economic case

Figure 4. **SMR key economic drivers to compensate for diseconomies of scale**

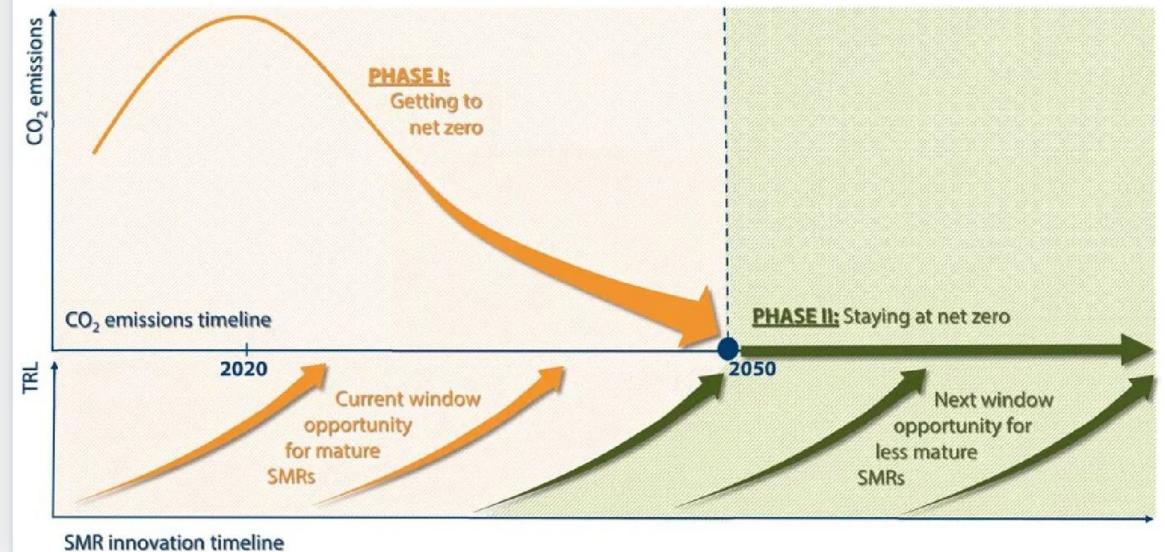


Source: NEA (2021b).

The **four drivers** that offset the diseconomy of scale

Figure 3. **Windows of opportunity for SMRs to support net zero objectives**

Near-term SMRs will contribute to net zero by 2050, with more innovative designs contributing to staying at net zero post-2050



Net-zero targets and the **time window** for SMRs

**diseconomy of scale** smaller output raising the cost per kW · **modularisation** building in a factory and shipping · **harmonisation** aligning licensing requirements across countries

Source This textbook §2.4.6, Figs. 2.19-2.20 · economy of scale  $C_2/C_1 = (P_2/P_1)^n$  with the exponent below 1

# Five different counts from the same dashboard

- One section alone carries five numbers - **98, 56, 74, 129, 79**. They look contradictory; all five are right
- §1.1's **“they count different things”** repeats here. Whichever you use, write down **what it counted and when**

| what it counted                                                | when                                         |
|----------------------------------------------------------------|----------------------------------------------|
| <b>98</b> Designs identified worldwide, including discontinued | 2nd Edition · 2024                           |
| <b>56</b> Of those, the ones the dashboard scored              | 2nd Edition · 2024                           |
| <b>74</b> The Third Edition report’s scope                     | 3rd Edition · Jul 2025 (cut-off 14 Feb 2025) |
| <b>129</b> Designs NEA tracks                                  | Digital dashboard · rolling                  |
| <b>79</b> Of those, the ones listed on the digital dashboard   | Digital dashboard · rolling                  |

# 03 Collecting with AI

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Not “let AI do the research” but “how far to let it go and where to cut it off.” The conclusion is in the first line - AI stops at chapter 3; chapter 4 is done by a person.

# What AI does well, and what it **cannot** do

- Getting a number across intact is not a capability gap but **how it works** - it picks the next token by probability
- Same for "saying it does not know." A blank is the right answer, but it **leans towards filling it in plausibly**

## what it does structurally well

### ① Pulling scattered things together

Turning five designs into one table takes a person half a day and the model a minute.

### ② Pointing to where to look

Asked where the primary source for i-SMR specs is, it names the ARIS Design Status Report.

## what it structurally cannot do

### ① Carrying a number across intact

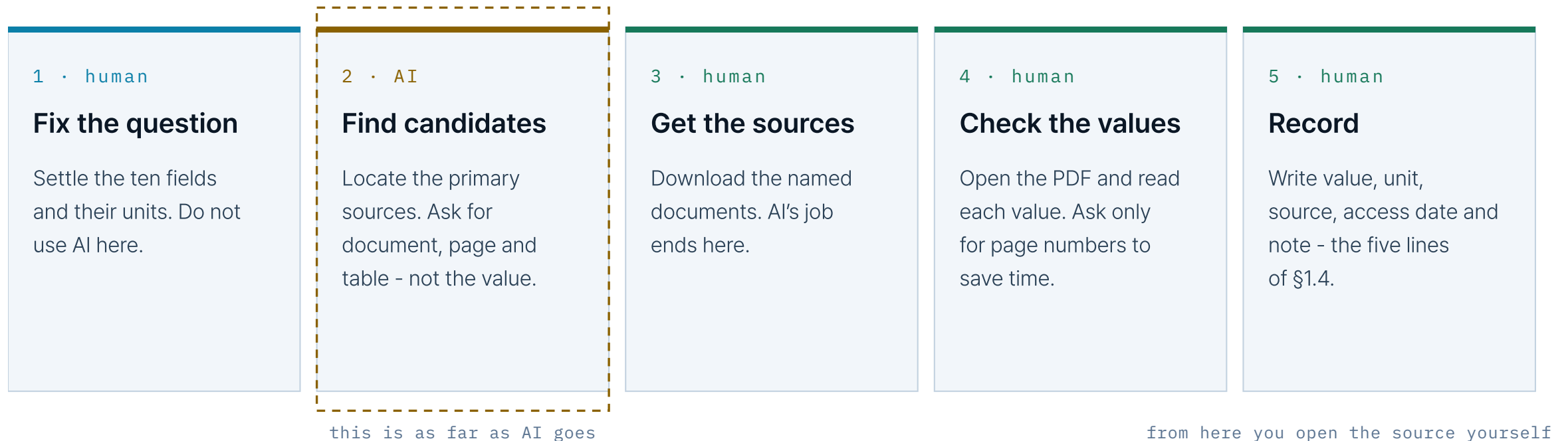
Reading a value out of a table into a sentence shifts rows and swaps units.

### ② Saying that it does not know

A blank is correct when the source is silent, but it leans towards a plausible fill.

# Five collection steps, and which one AI covers

- Not using AI in step 1 matters. You know what you need; let the model choose and the axis of comparison wobbles
- AI's job **ends at step 3**. Step 4 means opening the PDF and reading with your eyes - ask the model only for the page number



# Three kinds of AI collection tool

- A direct connection (API, MCP) has the model **query the database** instead of composing - what comes back is **a record, not a sentence**
- Building lab tooling? **Start with search and listing**. One tool that returns exactly what exists beats ten plausible summaries

## Deep research

what it does

Searches and reads on its own, then writes a report

where it fits

Step 2 - finding candidates

the risk

Citations appear, but the numbers drift

## Document Q&A

what it does

Answers only from the PDF you uploaded

where it fits

Steps 3-4 - reading the source

the risk

Misreads tables - merged cells, footnotes, units in headers

## Direct connection

what it does

Queries the database directly through an API or MCP

where it fits

Steps 2-3 - automated collection

the risk

Implementation cost and access rights

**MCP** Model Context Protocol, a standard for calling external tools and data · **API** a programmatic query channel into a database · **ADAMS** the NRC document system

**Source** This textbook §3.2, Table 3.1 · the landscape as of September 2026

# Writing the prompt - four rules

- Without it the model **prefers filling a blank to leaving it**. And a blank is itself information
- A good prompt does four things - **fix the fields, limit the source tier, allow blanks, block the known trap**

the bad one

Compare NuScale and i-SMR

- What to compare is missing
- As of when is missing
- In what units is missing

So the model replays a generic SMR comparison paragraph from training. Plausible, unsourced, and usually years out of date.

the good one - four rules

## ① Fix the fields

thermal, electrical, module count, pressure, inlet/outlet, flow, SG, pumps, fuel, design life

## ② Limit the source tier

ARIS reports, NRC documents and official vendor specs only. No news articles.

## ③ Allow blanks

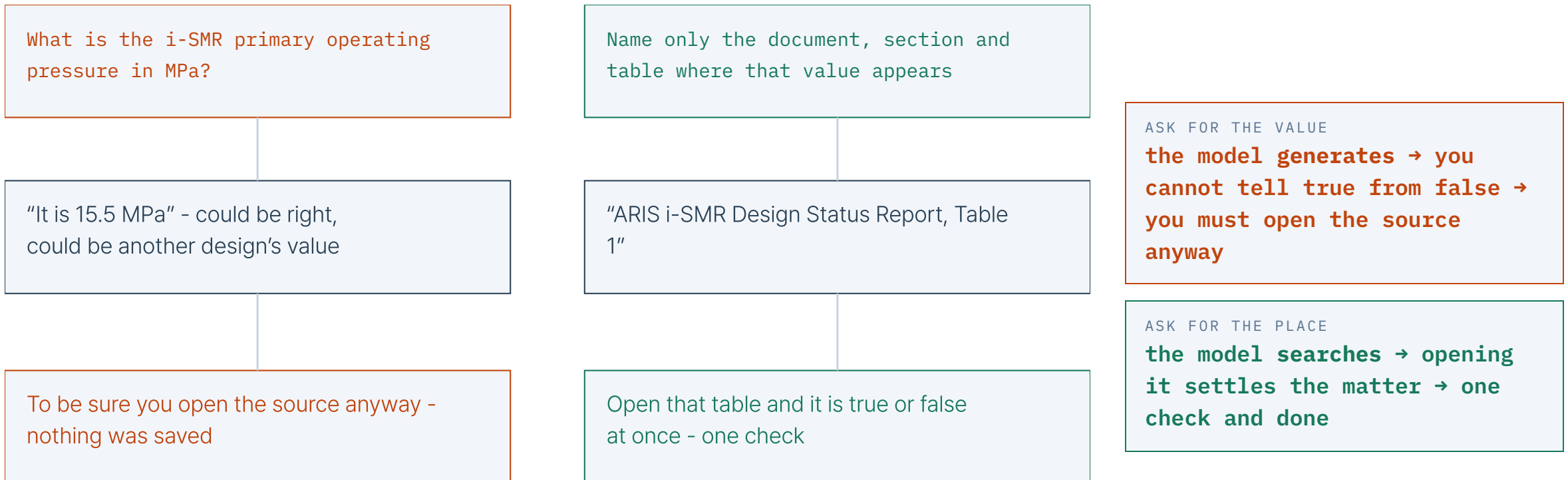
Leave unconfirmed fields blank and write "not confirmed." Do not fill in estimates.

## ④ Block the known trap

Do not mix US600 (160 MWt) material with US460 (250 MWt) material.

# Asking for a value versus asking for its location

- The two prompts differ by one line but the **cost of checking** is entirely different - the most practical trick in this chapter



# 04 Cross-checking

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Not “do not trust the tool” but narrowing down where to be suspicious. And the checking tool is one a thermal-hydraulics student already owns - one line of energy conservation.

# Three layers of a deep-research citation

- Deep-research output is dense with footnotes and links, and the links **open**. That is why it feels trustworthy
- Yet **23-61%** of cited facts are not actually supported by the source. Opening is a different question from supporting

## Does the link work?

It opens when you click it



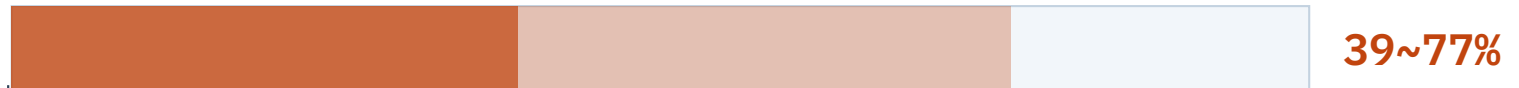
## Is the content relevant?

The document is at least about the topic



## Does it support the claim?

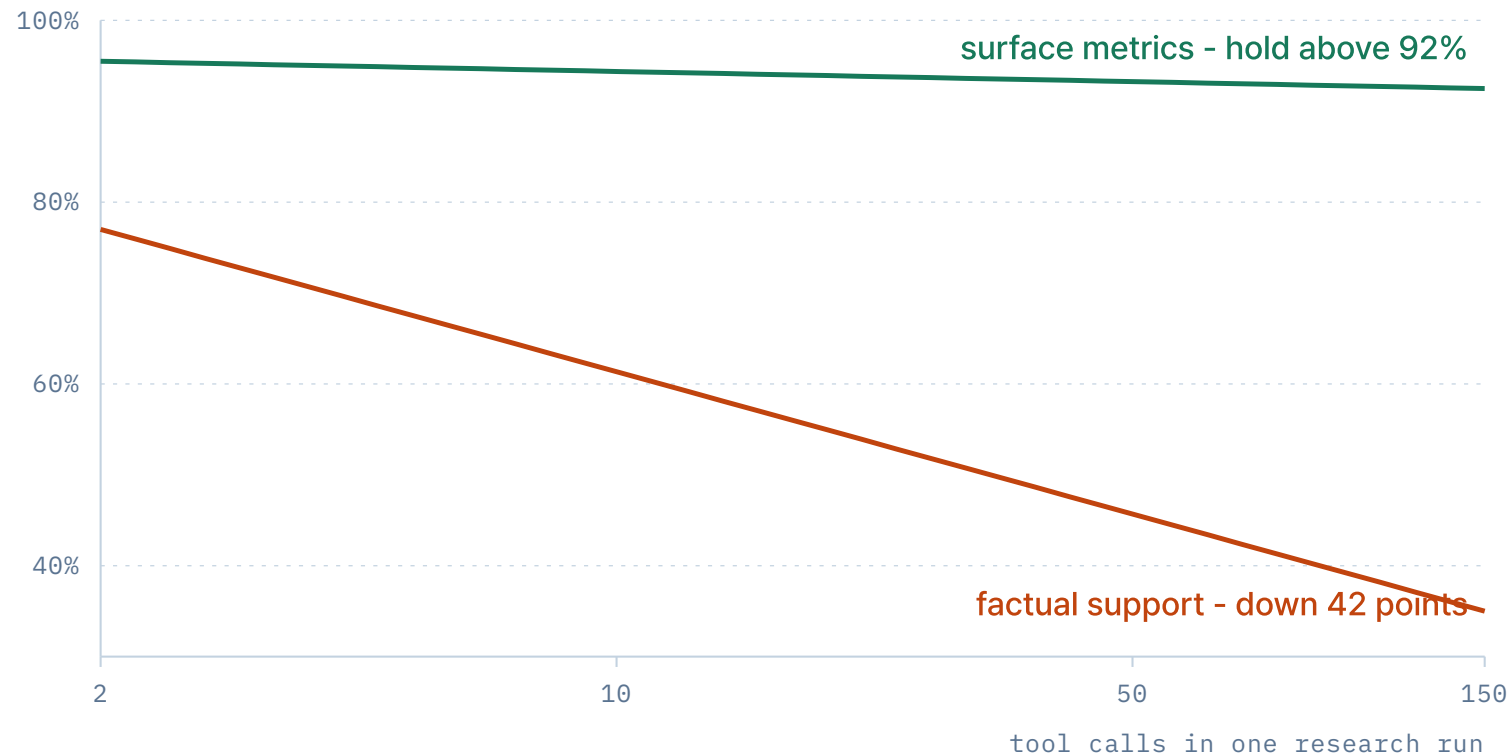
This is where it breaks - 23-61% of citations are unsupported



Top models. The solid band is the best value, the faint band the lower bound.

# Research depth against factual accuracy

- Raise tool calls from 2 to 150 and factual support drops **42 points on average** - while the surface metrics **hold above 92%**
- Split the question instead of widening it. Asking two or three fields at a time beats asking ten at once



Told to search longer, the report came back **longer, more plausible and more wrong**. A wider search means more material, and pairing each sentence with the right source gets harder.

*Dangerous for a reviewer - the visible signals are unchanged, so the eye cannot tell*

# Four types of numerical error, and how to catch each

- ③ Row slip matters most. A wrong value does not just sit there wrong - it **stops matching its neighbours physically**

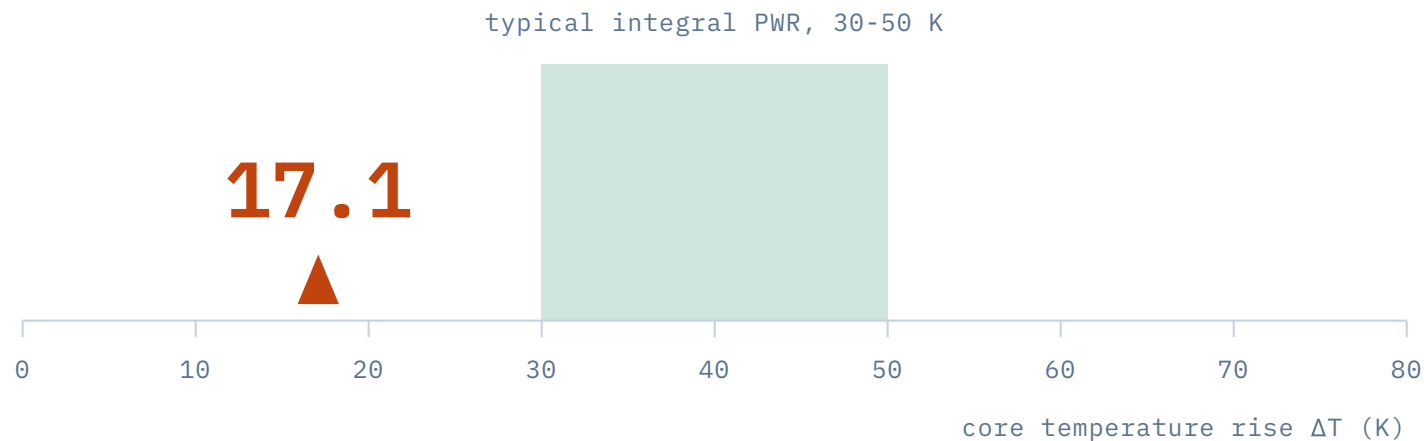
| symptom |                                                                                       | how to catch it                                                |
|---------|---------------------------------------------------------------------------------------|----------------------------------------------------------------|
| ①       | <b>Generation mix-up</b><br>Mixes an older and newer design of the same reactor       | Nail the design name (US600 / US460) into the prompt           |
| ②       | <b>Basis mix-up</b><br>Mixes per-module with per-plant, gross with net                | Write the unit and the basis into the field name itself        |
| ③       | <b>Row slip</b><br>Picks up the value from the next row or column                     | Read the neighbouring values too and test them for consistency |
| ④       | <b>Misattribution</b><br>Attaches a value to a real document that does not contain it | Demand the page and table number, then open that page          |

US600 / US460 NuScale’s earlier and later designs, 160 and 250 MWt · **gross** output at the generator terminals · **net** output after in-house consumption

Source This textbook §4.2, Table 4.1 · each is caught differently

# Catching a row slip with an energy balance

- Got two or more numbers? **Multiply and divide them.** A slipped row cannot satisfy the physics, so it **falls out before you open the source**



Outside the usual band - one of the two belongs to another design

$$Q = \dot{m} c_p \Delta T$$

THERMAL POWER  $Q$  250 MWt

PRIMARY FLOW  $\dot{m}$  2,660 kg/s

CORE TEMPERATURE RISE  $\Delta T$

17.1 K

the slipped pair

i-SMR as published

$Q$  core thermal power ·  $\dot{m}$  primary flow ·  $c_p$  specific heat at constant pressure ·  $\Delta T$  core temperature rise · **typical integral PWR 30-50 K**

Source This textbook §§4.2 and 4.4 · an estimate taking  $c_p \approx 5.5 \text{ kJ}/(\text{kg} \cdot \text{K})$  for water near 15.5 MPa and 300 °C

# The three-pass check, with two worked cases

- If pass 3 disagrees, **record the disagreement**. Silently picking one side is the most dangerous move

## Pass 1 - self-consistency

Do the values agree physically? The cheapest pass - no source needed.

## Pass 2 - open the source

Open that page of that document - the PDF, not the summary. Misattribution dies here.

## Pass 3 - an independent source

A second primary source by a different body. Two copies of one press release are not independent.

### Case A · i-SMR thermal power

**520 MWth vs 540 MW(th)**

Pass 1 cannot separate them - both give a thermal efficiency inside the usual band.

In pass 2 only ARIS supplies flow and temperatures, and they check out (512 MW, within 5%). Cite 540 and footnote the 520.

### Case B · NuScale operating pressure

**12.8 MPa vs 13.8 MPa**

Both are right. Different designs - 12.8 is US600 (160 MWt), 13.8 is US460 (250 MWt).

The source says 2 000 psia. Always suspect the point where the unit system changes - the rounding is the transcriber's.

**self-consistency** whether the values agree with each other physically · **independent source** two primary sources written by different bodies ·

**psia** pounds per square inch absolute - 2 000 psia = 13.8 MPa

**Source** This textbook §§4.3-4.5 · i-SMR 520 vs 540 MWth, NuScale 12.8 vs 13.8 MPa - both met in practice

# Verification **checklist** - six items

- All of Part I compresses into these six. Tick all six and the table **survives the question in the room**

- |                                     |                                                                              |      |
|-------------------------------------|------------------------------------------------------------------------------|------|
| <input checked="" type="checkbox"/> | Every value carries a document name and a page or table number               | §1.4 |
| <input checked="" type="checkbox"/> | Every source is primary - not an article, not a summary                      | §1.2 |
| <input checked="" type="checkbox"/> | Values of one design agree physically - the energy balance closes            | §4.2 |
| <input checked="" type="checkbox"/> | No mixing of design generations or bases (per module vs plant, gross vs net) | §4.5 |
| <input checked="" type="checkbox"/> | Where sources disagree, the disagreement is written down                     | §4.4 |
| <input checked="" type="checkbox"/> | The access date is recorded                                                  | §2.4 |

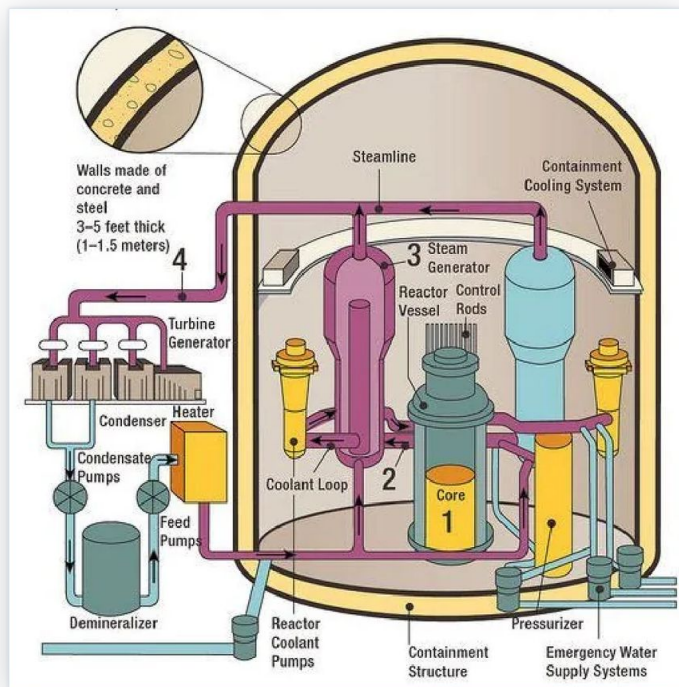
# 05 **PWR, BWR, and going integral**

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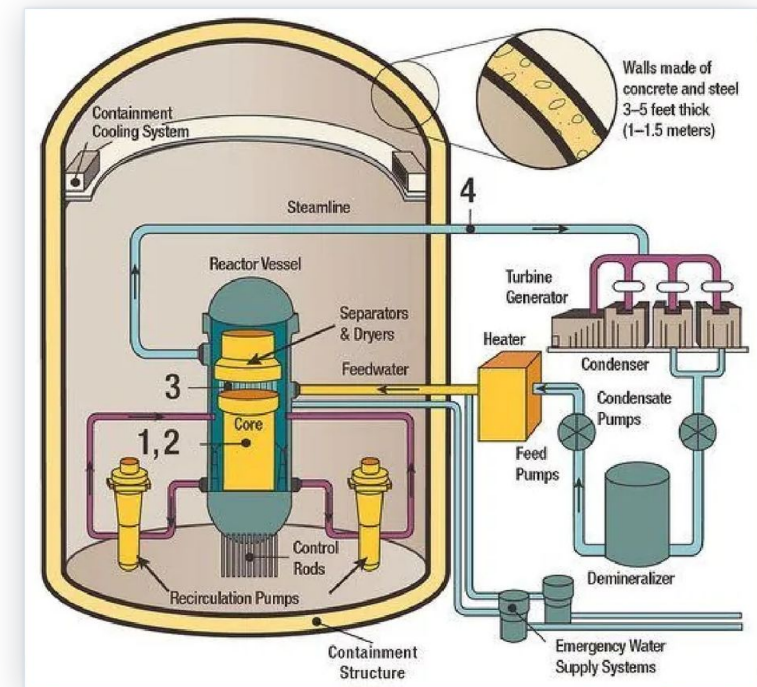
Where Part II starts. The two light-water lines split on one question - where the steam is made. Then: why SMRs converged on the integral layout.

# PWR and BWR - where the steam is made

- A **PWR** holds the primary at high pressure so **the core never boils**; heat crosses to the secondary in a steam generator. The point is **separation**
- A **BWR** boils in the core and sends that steam to the turbine. No steam generator - and **the turbine hall becomes a radiation area**



**PWR** - the primary (core to steam generator) and secondary (steam generator to turbine) are separate



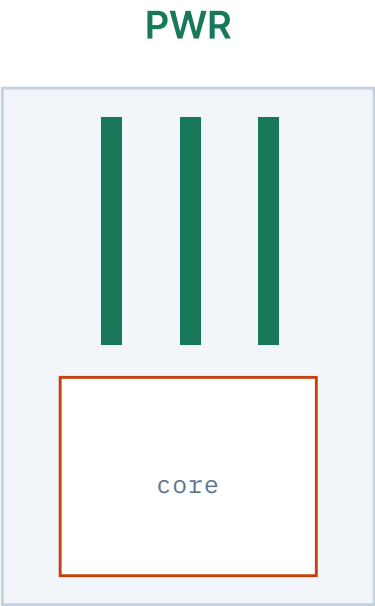
**BWR** - the steam-water mixture from the core passes the separator and dryer straight to the turbine

**PWR** Pressurized Water Reactor · **BWR** Boiling Water Reactor · **Rankine cycle** the steam power cycle · **steam separator** splits steam from water

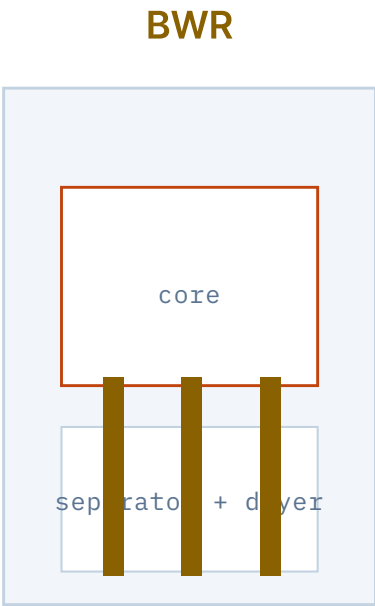
**Source** This textbook §5.1, Figs. 5.1-5.2 and Table 5.1 · representative pressures: PWR about 15.5 MPa, BWR about 7 MPa

# Rod insertion direction and behaviour on loss of power

- A BWR's core is capped by separators and dryers, so rods must be **driven in from below** - the structure forces it
- So a BWR needs **hydraulic and electric drives with accumulators**. It comes back often in safety analysis



Cut the power and they drop - cutting it is the action



They do not insert on their own - a separate hydraulic accumulator is needed

| ITEM            | PWR                | BWR                |
|-----------------|--------------------|--------------------|
| Steam made in   | Steam generator    | The core           |
| Cycle           | Two loops          | One loop           |
| Pressure        | 15.5 MPa           | 7 MPa              |
| Steam generator | Yes                | No                 |
| Rod insertion   | Top down · gravity | Bottom up · driven |
| Turbine hall    | Low                | High               |

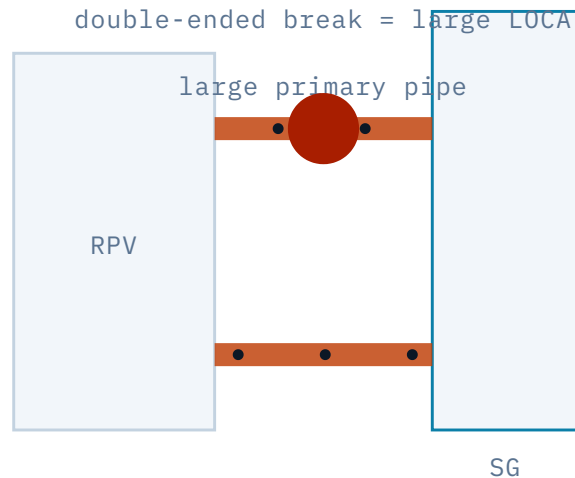
**control rod** absorbs neutrons to cut reactivity · **accumulator** a vessel storing force in compressed gas · **FMCRD** Fine Motion Control Rod Drive (BWRX-300)

**Source** This textbook §5.1.2, Table 5.1 · remember this when the BWRX-300 FMCRD comes up in ch. 9

# The integral layout - removing the large-LOCA scenario

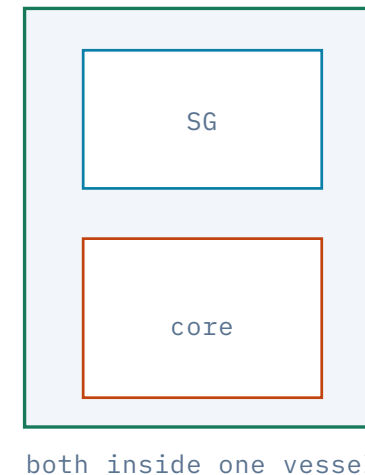
- Safety injection, accumulators and recirculation in a large PWR all exist for **one scenario** - a large pipe break
- No pipe, no scenario. **Zero break area means zero escaping flow**

## Loop layout - the pipes are there



Injection, accumulators and recirculation all exist for this one case

## Integral - no pipes at all



$$A_{\text{break}} \rightarrow 0 \Rightarrow \dot{m}_{\text{loss}} \rightarrow 0$$

What remains - instrument penetrations, feed and steam nozzles, relief valves

**LOCA** Loss-of-Coolant Accident ·  $A_{\text{break}}$  break area ·  $\dot{m}_{\text{loss}}$  escaping flow · **accumulator** a passive tank that pushes water in with compressed gas

**Source** This textbook §5.2 · a large LOCA is the double-ended rupture of a large primary pipe

# What the integral layout **gains and gives up**

- A LOCA does not vanish entirely. The vessel still carries **penetrations, nozzles and relief valves**

what it gains

The large-break scenario is removed

The safety systems get simpler

Fewer systems, so a smaller site

Smaller radiation areas

what it loses

The vessel grows larger and thicker

The steam generator must be serviced inside the vessel

Harder to fabricate and to transport

Output is tied to how big the vessel can be

# Three designs, by how the coolant circulates

- Even after going integral one fork remains - **what drives the coolant round**
- The three **answer one question differently**. Not better or worse - a difference in what was optimised

## How do we deal with a pipe break?

ch. 6

### Drop the pump

NuScale VOYGR

Height and density difference drive it

gains

No moving parts left

the price

The vessel has to get longer

ch. 7

### Keep the pump

i-SMR

Bolted to the vessel, no pipe

gains

Flow becomes a design variable

the price

Power and rotating machinery stay

ch. 9

### Boil in the core

BWRX-300

Void buoyancy drives it

gains

No steam generator at all

the price

Pipes remain; isolation handles them

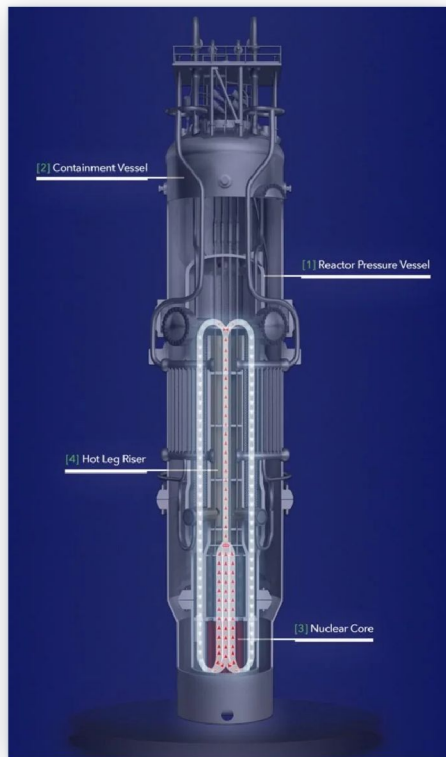
# 06 NuScale VOYGR

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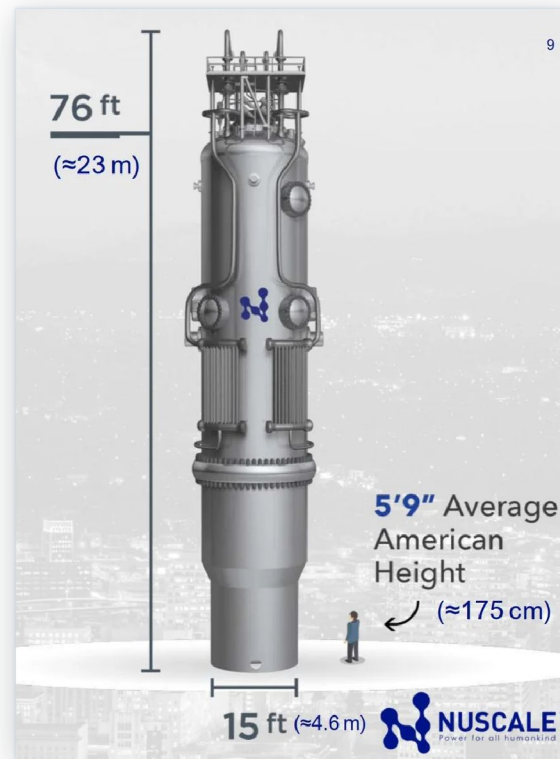
The first SMR to win a design approval in the United States. Reactor, steam generator, pressuriser and even the containment in one cylinder. We start inside the vessel and take the safety systems apart one by one.

# The NuScale Power Module - layout and dimensions

- The outer cylinder is the **containment**, the inner one the **reactor vessel**. The long central tube is the riser, with the helical steam generator wound in the annulus beside it
- Set beside a person it shows: this is **a transportable size**. The whole factory-build premise follows from these dimensions



NPM section



23 m tall, 4.6 m across



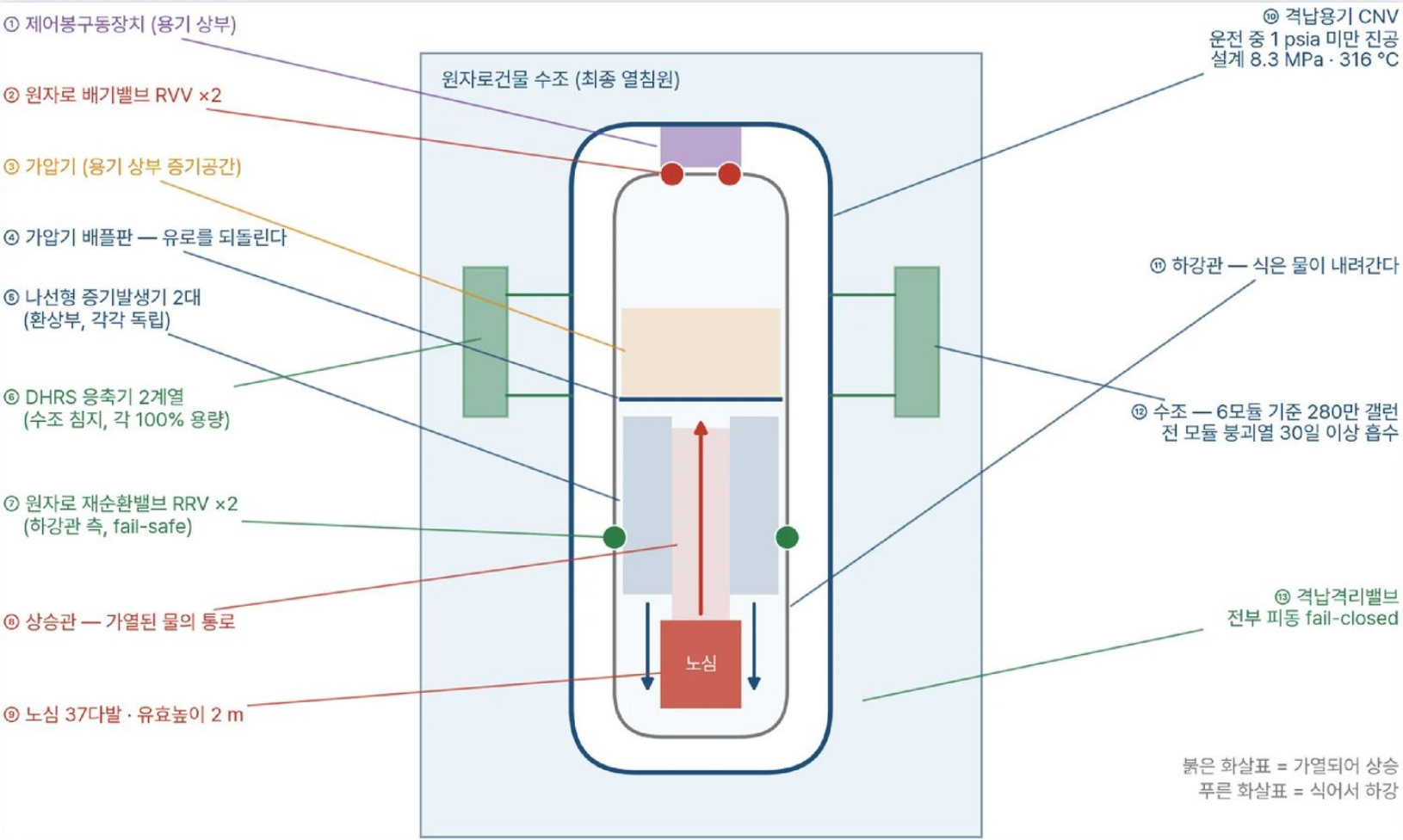
In the factory

NPM NuScale Power Module · **containment** the pressure boundary that confines radioactivity · **riser** the passage the heated water climbs

Source This textbook §6.1, Figs. 6.1-6.2 · module 76 ft (about 23 m) tall and 15 ft (about 4.6 m) across

# Components inside the module

- The reactor vessel and containment are **nested**, and both sit **submerged in a pool**. That one fact drives every safety argument that follows

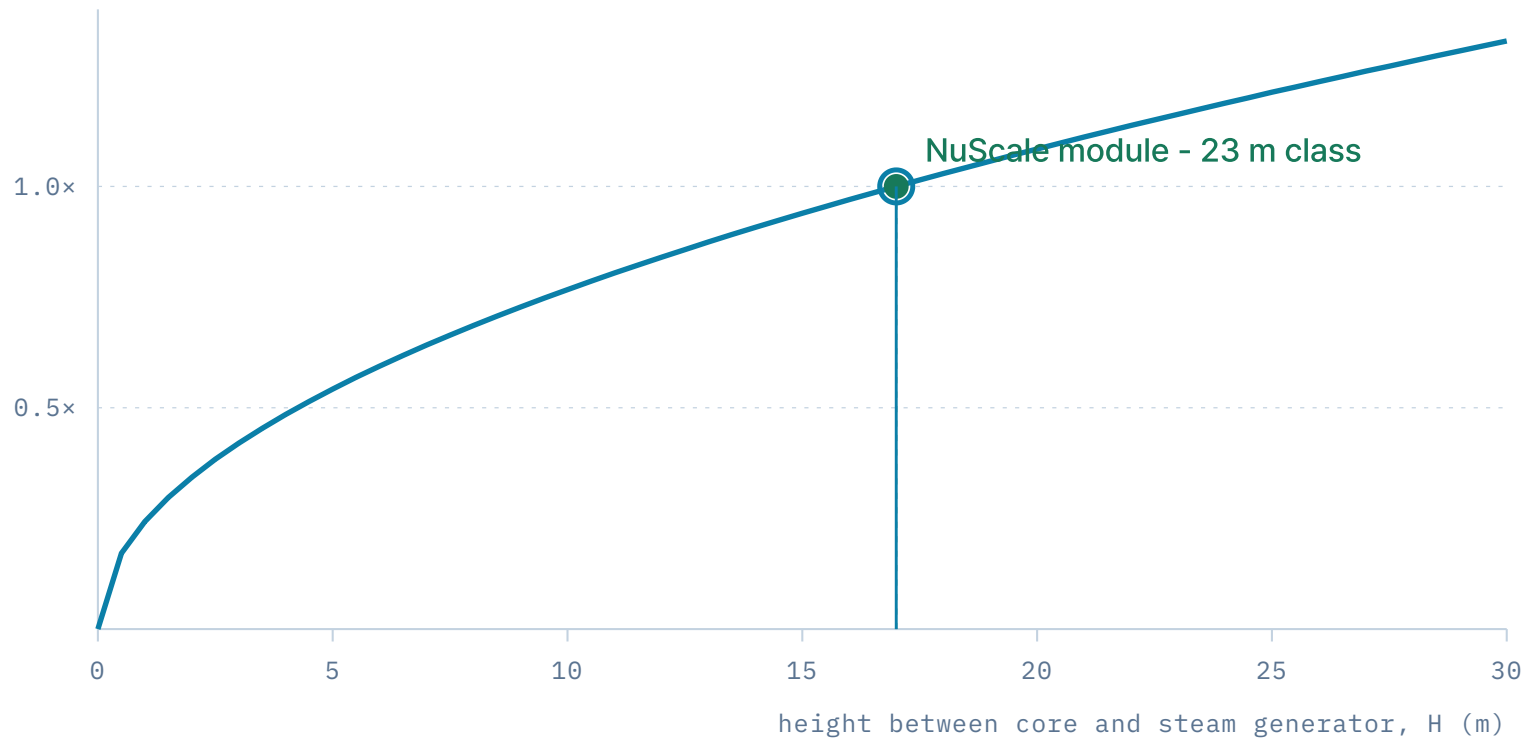


|                     |                                                |
|---------------------|------------------------------------------------|
| CRDM                | upper head; gravity drop on loss of power      |
| Pressuriser         | steam space at the very top                    |
| RVV ×2              | pressuriser region; vents steam to containment |
| Steam generators ×2 | in the annulus; two independent units          |
| Riser               | centre, above the core; source of the head     |
| Downcomer           | between vessel wall and riser                  |
| RRV ×2              | vessel side, at downcomer level                |
| Core                | 37 assemblies, about 2 m active                |
| CNV                 | pressure boundary and condensing surface       |
| Pool                | heat sink, shielding, transport route          |
| CIV                 | all passive, fail-closed                       |

The NuScale module component by component - two nested vessels, both submerged

# Natural-circulation flow and vessel height

- Drop the pump and buoyancy drives the loop. Flow goes as **the square root of the height** - hence the long, slim shape
- Cutting the loss coefficient **K** matters as much as adding height - **this is where CFD actually earns its keep**



$$\dot{m} \propto \sqrt{(\Delta p \cdot H)}$$

HEIGHT H

17.0 m

RELATIVE FLOW

1.00 ×

HEIGHT FOR THE SAME FLOW

17.0 m

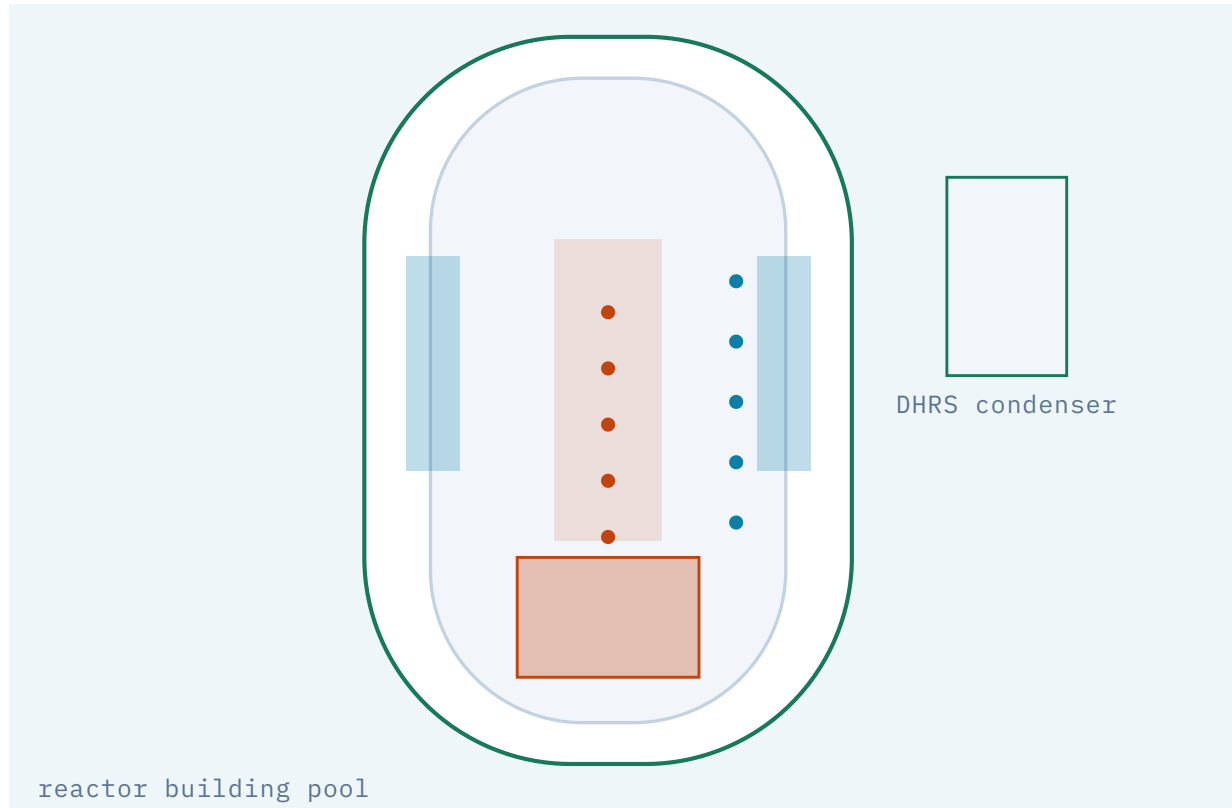
The module is 4.6 m across and 23 m tall.  
**The price of dropping the pump is paid in shape.**

**H** height between the centre of the core and the centre of the steam generator · **K** total loop loss coefficient ·  $\rho_c, \rho_h$  cold-side and hot-side density

**Source** This textbook §6.3 · flow settles where the driving head  $\Delta p = (\rho_c - \rho_h) g H$  equals the loop loss  $K \dot{m}^2 / 2\rho A^2$

# Flow path by mode - normal, DHRS, ECCS

- Normal → transient → accident: **only the path changes, inside the same vessel**. At no point is water pumped in from outside



A · Normal - natural circulation

B · Transient - DHRS boil and condense

C · Accident - ECCS boil and condense

NORMAL · NATURAL CIRCULATION

Heated in the core, up the riser, turned at the pressuriser baffle, down past the steam generator shell side, down the downcomer, back to the core. No pump anywhere.

**DHRS** Decay Heat Removal System, a closed loop on the secondary side · **ECCS** Emergency Core Cooling System · **RVV** reactor vent valve · **RRV** reactor recirculation valve

**Source** This textbook §6.4, Figs. 6.4-6.5 · (A) normal natural circulation, (B) DHRS, (C) ECCS

# The vacuum containment - why, and its design pressure

- A large PWR containment holds **0.4-0.5 MPa**. NuScale's holds **8.3 MPa**, because primary steam is vented into it **without depressurising first**
- It can be thick **because it is small**, and direct venting is possible **because it can be thick**

## ① It cuts parasitic heat loss

A vessel running submerged leaks heat from its surface all the time. Empty the gap and the conduction and convection paths are gone - the physics of a vacuum flask.

## ② Nothing is left to block condensation

Non-condensables form a blanket on the condensing surface and cut heat transfer badly. A large PWR containment is full of air and is designed around that; here the air was never there.

## ③ The corrosion environment improves

With no oxygen and no moisture, corrosion of the outer vessel surface is much less of a worry.

containment design pressure

Large PWR containment



0.4~0.5 MPa

NuScale containment (CNV)



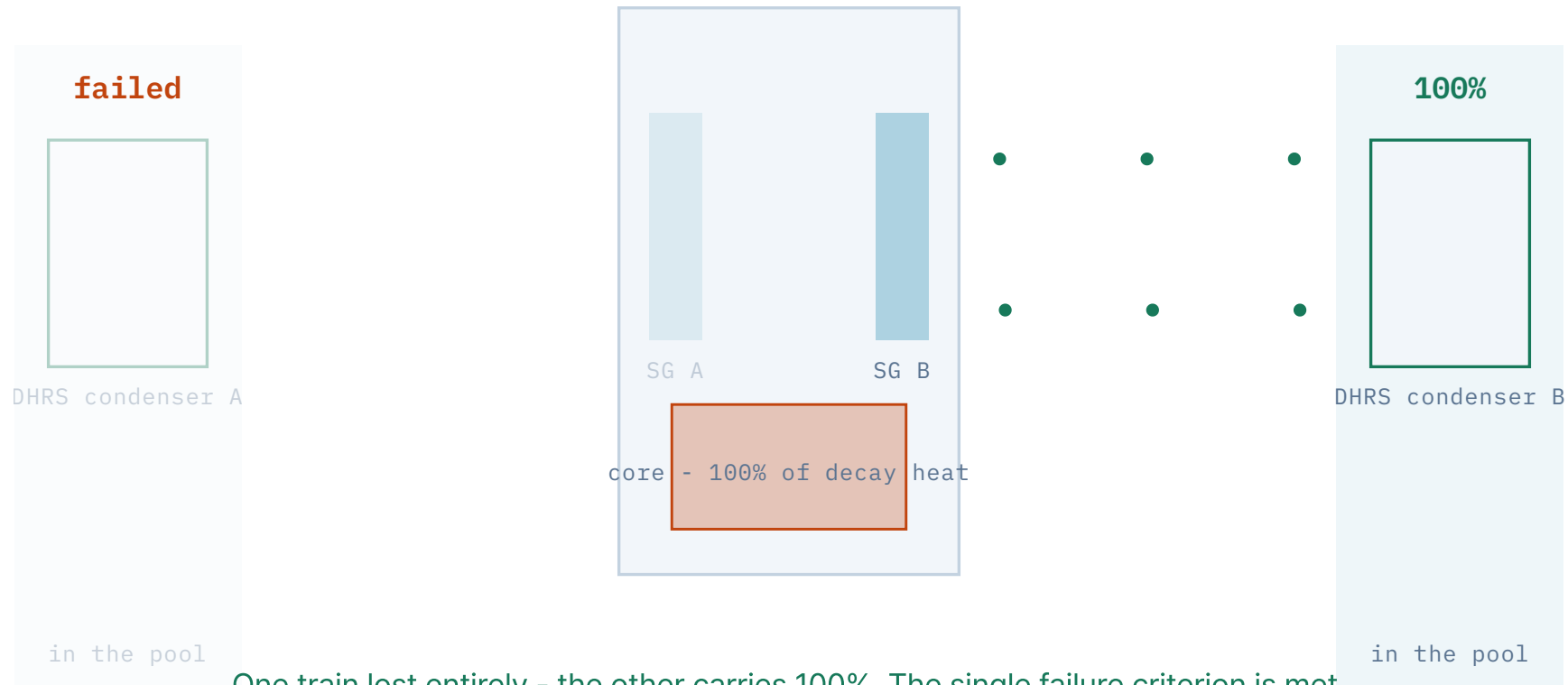
**non-condensable gas** a gas such as air that stays gaseous while steam condenses · **condensation heat transfer** steam turning to water on a cold surface and giving up its heat

**Source** This textbook §6.2.1 · containment held **below 1 psia (about 6.9 kPa)** in operation; design pressure 8.3 MPa (1 200 psi)

# DHRS - two trains, each rated 100%

- One train per steam generator, and **each train alone removes the full decay heat** - lose a whole train and the function stands
- Once the valves open, **thermodynamics runs it**. No electricity, no operator action

Two trains, one per steam generator, each rated for 100%



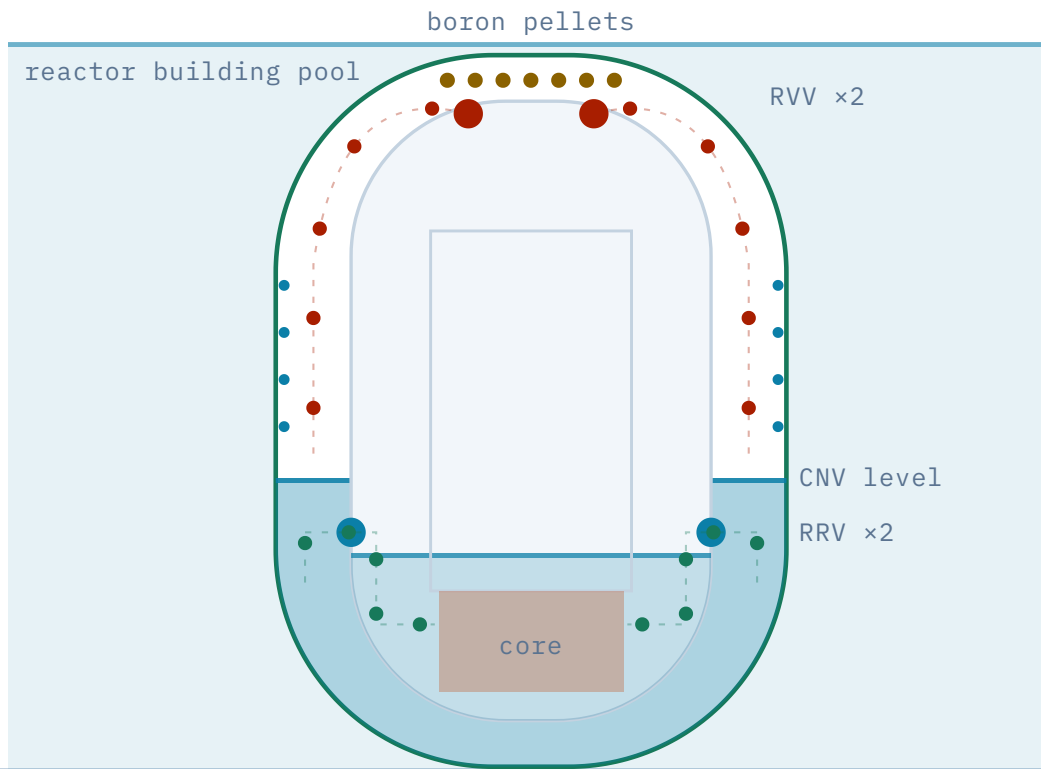
One train lost entirely - the other carries 100%. The single failure criterion is met.

**DHRS** Decay Heat Removal System · **single failure criterion** the function must survive any one component failing · **2 × 100%** either train alone carries the full duty

**Source** This textbook §6.5.1, Table 6.2 · each train alone can remove the full decay heat

# ECCS - venting through RVVs, returning through RRVs

- A large PWR's ECCS pumps water in, so it needs **pumps, power and a water source**. Here **none of the three is needed**
- When the ECCS acts, **boron pellets in the upper containment dissolve** and boron goes in with the water - it melts on contact, needing no signal and no power



## 1 The RVVs open

Steam from the top of the vessel goes into the containment

## 2 It condenses on the wall

The containment is submerged, so the wall is the heat exchanger

## 3 Boron goes in with it

Pellets in the upper containment dissolve and raise the primary boron - the second means under GDC 26

## 4 The RRVs return it

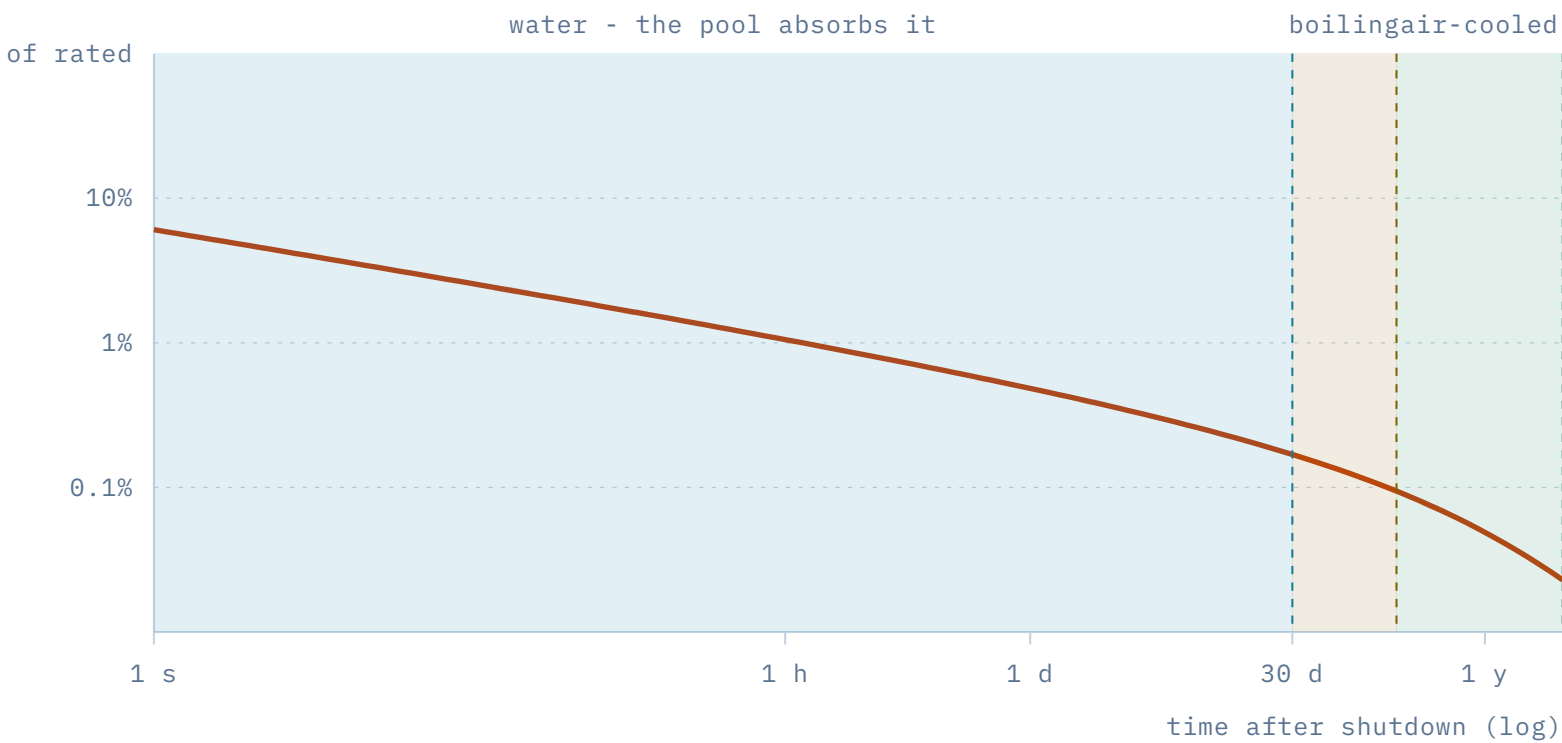
Once the level covers the RRVs, water returns to the downcomer - no pump, no power, no extra source

IAB Inadvertent Actuation Block · GDC 26 the US criterion requiring two independent reactivity control systems · boron pellet a solid that dissolves on contact with water and releases boron

Source This textbook §§6.5.2-6.5.3 · each valve has a main valve, an inadvertent actuation block and two solenoid pilots (actuate and reset)

# The ultimate heat sink - pool, boiling, air

- Air cooling cannot carry the decay heat right after shutdown. **Water → boiling → air**, in that order and at those times, is the design
- Water inventory per MW is about **four times** a conventional PWR's. Four times the denominator means **a quarter of the temperature rise rate** - that one relation sets the coping time



$$dT/dt = Q_{decay} / M c_p$$

|                          |                     |
|--------------------------|---------------------|
| Pool (6 modules)         | about 2.8 M gal     |
| Full decay heat absorbed | 30+ days            |
| After 30 days            | evaporation + air   |
| ELAP analysis            | verified to 99 days |

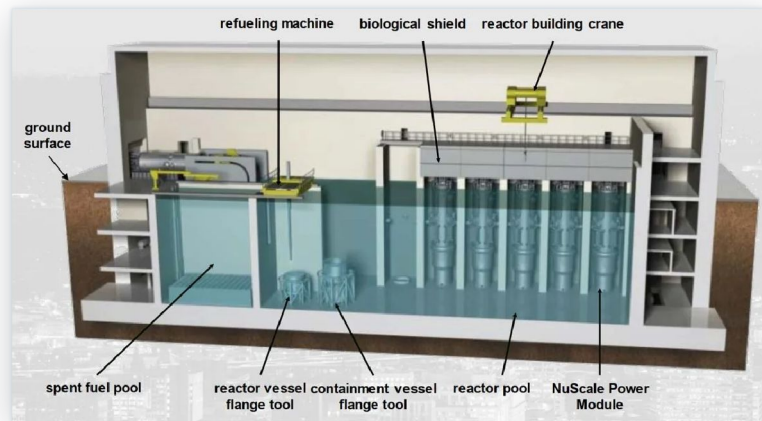
When you show this, **always show the time axis with it**. Quoting the final state alone gives a false impression.

No pump, no external power, no external water - it moves through the three in turn  
ELAP Extended Loss of AC Power, a beyond-design-basis scenario · **decay heat** heat from fission products after shutdown · **ultimate heat sink** where the heat finally goes

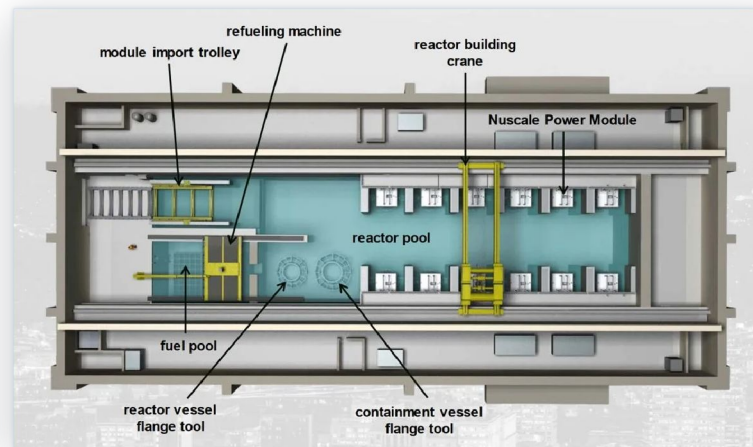
Source This textbook §§6.5.5 (Table 6.3) and 6.6 · about 2.8 million gallons for six modules; the ELAP analysis keeps water above the core to 99 days

# The reactor building and site layout

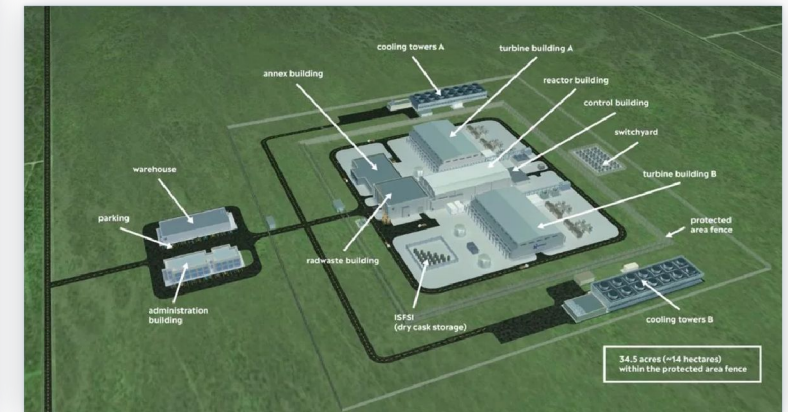
- The modules sit **side by side in a shared pool** with a transfer crane above. Servicing one means **lifting and moving it inside the pool**
- Reactor, turbine and control buildings and the cooling towers all fit on **a smaller footprint than a large plant** - a consequence of having fewer systems



Building section



Building plan



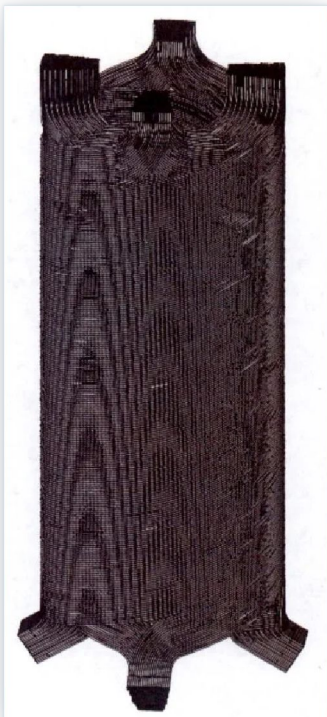
Site layout

ultimate heat sink where the heat finally goes · shielding what stops the radiation · spent fuel pool the tank where discharged fuel cools and is stored

Source This textbook §6.7, Figs. 6.8-6.10

# The helical once-through steam generator

- In an integral design the steam generator **must fit inside the vessel**, so volume is the binding constraint - and that is where helical wins
- Once-through boiling inside the tubes, **inclined cross-flow** outside. Where they meet is **the hard problem of integral SMR analysis**



The helical coil



The real thing - two independent bundles

The thermal-hydraulics of helical steam generators is a **THINK Lab topic on the i-SMR project**. The flow outside the bundle is an **inclined cross-flow**, and once the bundle is large you cannot model tube by tube - hence the **porous-medium approximation**.

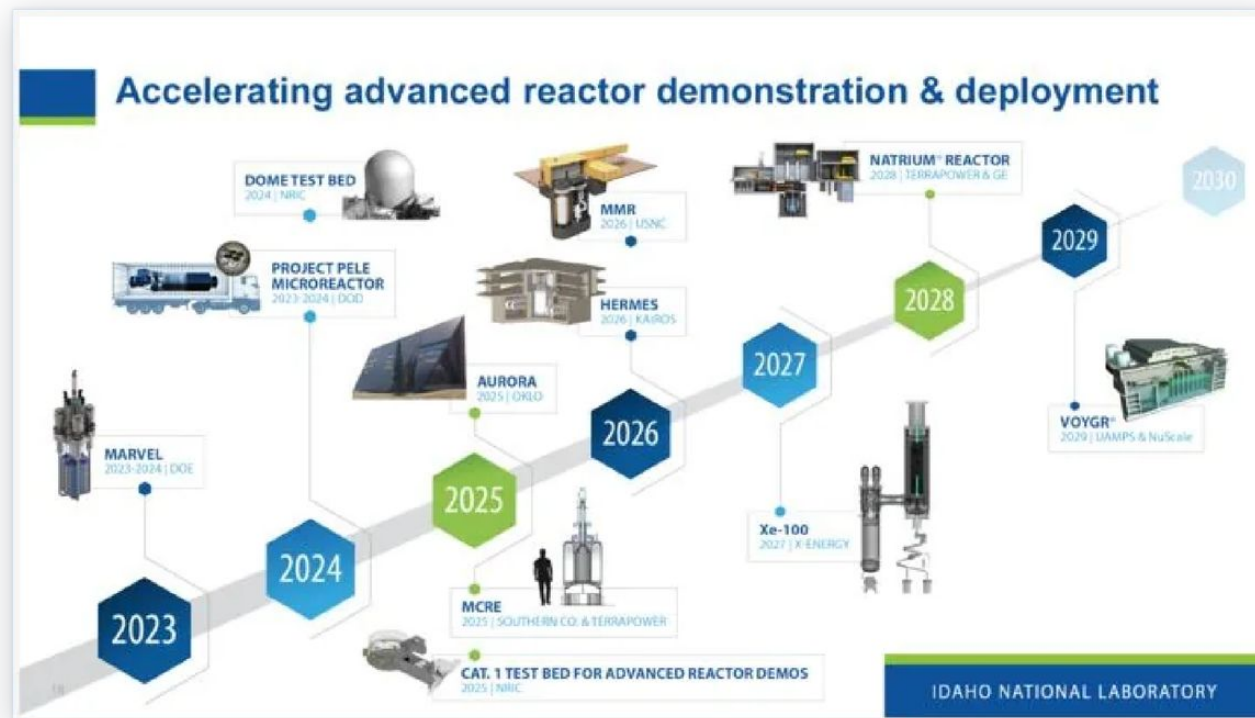
ATHOS-class codes are used for this

**once-through** water becomes steam in a single pass · **shell side** outside the tube bundle · **porous-medium approximation** modelling the bundle as a block

**Source** This textbook §6.8, Figs. 6.11-6.12 · primary on the shell side, secondary inside the tubes - a **once-through** design

# Factory fabrication and the emergency planning zone

- The claim is that a small core and deep defence let the EPZ shrink **to the site boundary**, supported by source-term and release analyses
- If it shrinks, **siting next to an industrial park becomes possible** - which is what opens up process heat and hydrogen



The advanced-reactor demonstration schedule at Idaho National Laboratory - several designs phased onto one site

EPZ Emergency Planning Zone, where evacuation and sheltering plans must exist · **source term** how much of what could be released · **INL** Idaho National Laboratory

**Source** This textbook §6.9 (Fig. 6.14) and §6.9.1 · the EPZ case rests on **source term and release analyses**

EPZ OF A LARGE PLANT

**about 16** km

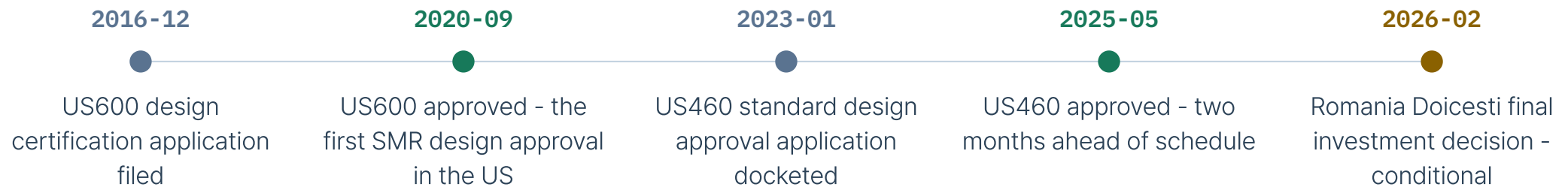
EPZ NUSCALE ARGUES FOR

**the site boundary**

Site construction becomes **fabricate, ship, install**. The schedule shortens and quality moves into the factory - but **transport limits become a design constraint**.

# NuScale **licensing and project record**, September 2026

- Conditional FIDs are common and the conditions are usually **not published**. Blur them together and you **overstate the progress**
- In a licensing table always separate **pre-application, design approval, construction permit, FID and first concrete**



## What a design approval guarantees

That the regulator reviewed and accepted the design's safety. It says nothing about where or when anything gets built.

## What an FID guarantees

That the owner has decided to invest. It is often conditional, the conditions are usually private, and it is not a construction start.

# 07 i-SMR

---

Korea's innovative SMR. Integral like NuScale, but different in circulation, reactivity control and containment. Its specifications come from the Design Status Report lodged with IAEA ARIS.

# The i-SMR module - layout and flow path

- Up through the core, inside the core support barrel and upper guide structure, turned at the top, through **the coolant pumps**, down past the steam generator shell side to the lower plenum, and back into the core
- Against NuScale the **structure is the same and only the driver differs**: both rise in the middle and fall on the outside, with the steam generator on the descending leg

① 가압기 (용기 상부)  
② 비상감압밸브 EDV × 2  
③ 내장형 제어봉구동장치 코일·모터·연장축 모두 용기 안  
④ 무밀봉 캔드모터 펌프 4대 (용기 상부 직결)  
⑤ 나선형 관류 증기발생기 1대 하강관 안·8구역 분할  
⑥ PCCS 열교환기 24시간 내 격납압력을 절반으로  
⑦ 비상재순환밸브 ERV × 2  
⑧ 노심 69다발 17×17 가연성흡수체  $Gd_2O_3-^{235}UO_2$   
⑨ 강제 격납용기 설계압력 5 MPa  
⑩ 유로: 노심 → CSB-UGS 내부 → RCP → SG 셀측 하강 → 하부플레넘  
⑪ 격납 외벽 직결 이중 격리밸브  
⑫ ECT — 고갈 후 격납 외부 공기 자연순환으로 전환  
EDV로 방출 → PCCS가 응축 → 압력 평형 후 ERV로 재순환

|                 |                                                 |
|-----------------|-------------------------------------------------|
| PZR             | top of the vessel; holds pressure               |
| EDV × 2         | upper vessel; releases to containment           |
| CRDM            | internal; coils and motor all inside            |
| RCP × 4         | upper vessel; sealless canned motor             |
| Steam generator | helical once-through, in the downcomer, 8 zones |
| PCCS            | outside containment; condenses its steam        |
| ERV × 2         | lower vessel side; returns condensate           |
| Core            | 17×17, 69 assemblies                            |
| Containment     | small steel, 5 MPa design                       |
| CSB · UGS       | forms the riser, supports the core              |

RCP re  
· UGS

Source This textbook §§7.1-7.2, Fig. 7.1 and Table 7.1 · IAEA ARIS, I-SMR Design Status Report

# Three choices that cut the penetrations

- All three choices fold into **one policy** - fewer holes through the pressure boundary

## ① The SG sits inside the downcomer

The heat transfer surface sits where the down-flow already passes. No separate shell space, so vessel volume is saved.

What it removes - vessel volume, and the wall thickness that goes with it

## ② The pumps are sealless canned motors

A large PWR coolant pump has a shaft seal, and seal failure is a path that probabilistic safety assessment cannot ignore.

What it removes - the seal LOCA path

## ③ The rod drives are inside the vessel

Coil stack, motor and extension shaft are all inside. Corrosion of upper-head penetrations has been a real problem in large PWRs.

What it removes - the upper-head penetrations

**seal LOCA** coolant escaping through a failed pump shaft seal · **canned motor** a pump whose shaft never leaves the pressure boundary · **penetration** a hole through the pressure boundary

**Source** This textbook §7.2.2 · count **the holes in the pressure boundary** and the character of a design shows itself

# i-SMR specifications - design values and targets

- Core damage frequency is **a value that probabilistic assessment must establish**, not one a spec sheet settles. Promotional material often loses the distinction

|                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>Power</div> <div>Rated thermal per module</div> <div>540 MW(th)</div> <div>Electrical per module</div> <div>170 MWe</div> <div>Reference plant</div> <div>4 modules · 680 MWe</div> <div>Cycle length</div> <div>24 months</div> | <div>Thermal-hydraulics</div> <div>Primary pressure</div> <div>15.5 MPa (abs)</div> <div>Core inlet / outlet</div> <div>286 / 321 °C</div> <div>Primary flow</div> <div>2 658 kg/s</div> <div>Steam generator</div> <div>helical once-through, 8 zones</div> | <div>Core and systems</div> <div>Fuel</div> <div>17×17 · 69 assemblies · under 5%</div> <div>Soluble boron</div> <div>not used</div> <div>Safety systems</div> <div>PAFS · PECCS · PCCS</div> <div>Unattended coping</div> <div>72 hours or more</div> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                        |           |                                |                    |                       |
|------------------------------------------------------------------------|-----------|--------------------------------|--------------------|-----------------------|
| Below are design targets, not design values - always cite them as such |           |                                |                    |                       |
| Design life                                                            | Seismic   | Core damage frequency          | Capital cost       | Build time per module |
| 80 years                                                               | 0.5 g SSE | below 1.0×10 <sup>-9</sup> /MY | below 3 500 \$/kWe | 24 months             |
|                                                                        |           | target                         | target             | target                |

safe shutdown earthquake

Source This textbook §7.3, Table 7.2 · IAEA ARIS, i-SMR Design Status Report · accessed 2026-09-16

# What forced circulation buys - $\Delta T = 35 \text{ K}$

- With forced circulation the flow is **an input**. So the core temperature rise can be **held near a large PWR's, about 35 K**
- Similar conditions mean **validated correlations, system codes and test databases carry over**. Choosing a design is choosing a validation burden

## Natural circulation (NuScale)

Flow is set by density and height difference

The designer cannot choose it -  $\Delta T$  follows

Low-velocity, low-drop regimes can fall outside existing correlations

→ separate validation tests become necessary

## Forced circulation (i-SMR)

Flow is an input - four pumps give 2 658 kg/s

$\Delta T$  was set at  $321 - 286 = 35 \text{ K}$

Almost the same as a large PWR at 30-35 K

→ validated correlations and codes carry over

the same relation checks the spec table (§4.4)

$$Q = \dot{m} c_p \Delta T = 2\,658 \times 5.5 \times (321 - 286) \approx 512 \text{ MW}$$

Within 5% of the rated 540 MWt - the ARIS values are self-consistent

**validation testing** confirming correlations and system performance by experiment

Source This textbook §7.4 ·  $Q = \dot{m} c_p \Delta T = 2\,658 \times 5.5 \times (321 - 286) \approx 512 \text{ MW}$ , within 5% of the rated 540 MWt

# The passive safety systems - PAFS, PECCS, PCCS

- The first 36 h is **cooling** (bring the temperature down); the next 36 h is **holding** (keep it there). Different requirements
- The first is a question of **heat removal capacity**, the second of **pool inventory**. The validation splits the same way

## PAFS

### Decay heat removal

On loss of normal feed, the secondary side condenses and returns by natural circulation. Validated by PASCAL, LAPLACE and ATLAS-PAFS.

## PECCS

### Core cooling and inventory

When the vessel-to-containment differential reaches its setpoint the EDVs open; after condensing, the ERVs recirculate. Pressure itself is the actuation signal, not an electronic one.

## PCCS

### Containment pressure and temperature

Condenses containment steam and dumps the heat to the emergency cooling tank. In a design-basis accident it halves the peak containment pressure within 24 h.

PAFS performance requirement - with no operator action

### first 36 h - cool it down

Remove decay heat and bring the plant to safe shutdown → a question of heat removal capacity

### next 36 h - hold it there

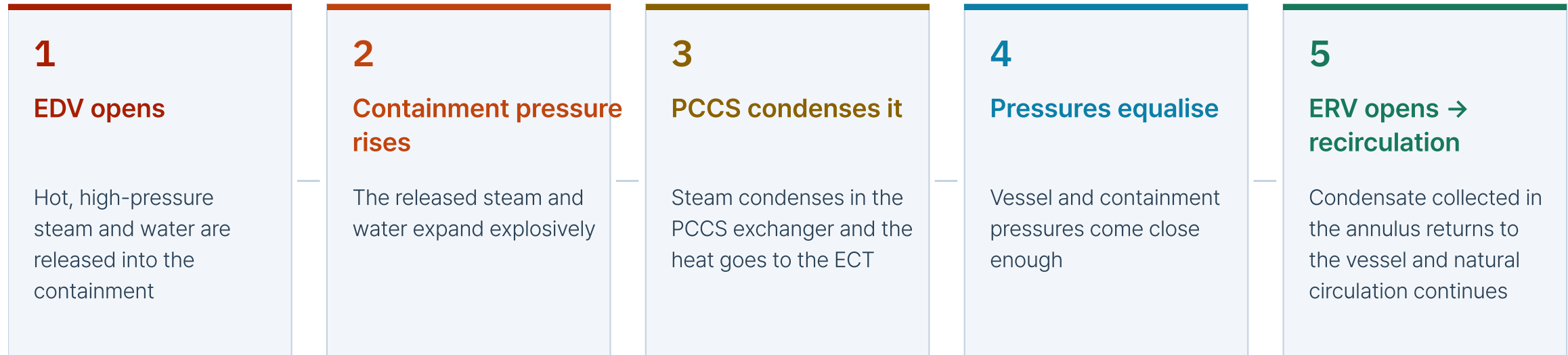
Hold that state for at least 36 more hours → a question of pool inventory

**PAFS** Passive Auxiliary Feedwater System · **PECCS** Passive Emergency Core Cooling System · **PCCS** Passive Containment Cooling System · **RSS** Reactor Safety System · **ECT** emergency cooling tank

**Source** This textbook §7.5, Table 7.3 · PAFS: 36 h to cool plus 36 h to hold · PCCS: half the peak pressure within 24 h

# The PECCS sequence - EDV opening to ERV recirculation

- Both are **depressurise** → **condense in the containment** → **return by gravity**. Nothing is pumped in
- In NuScale **the containment wall itself** condenses, because it is submerged; in i-SMR **the PCCS heat exchanger** does it



Step 1 does not end with opening a valve - THINK Lab's EDV dynamic load work

When the EDV opens, high-pressure saturated water flashes and chokes, and that puts dynamic loads on the piping and structures. A passive system is not an easier analysis.

**EDV** emergency depressurisation valve · **ERV** emergency recirculation valve · **flashing** saturated water turning to steam the instant pressure drops · **critical flow** the choked state where further depressurisation adds no flow

**Source** This textbook §7.5.2 · the actuation signal is **the vessel-to-containment differential pressure** reaching its setpoint, not an electronic trip

# Soluble-boron-free operation - gains and price

- A large PWR fills the gap with **boron injection** as its second independent means. NuScale's boron pellets sit in the same slot
- i-SMR must fill it with **rod worth and burnable absorber design** - a focus of the licensing review

what it gains

- ① The CVCS gets much simpler - injection, dilution and recovery equipment all go
- ② Less corrosion trouble - no boric acid precipitation or concentration
- ③ The moderator temperature coefficient stays reliably negative
- ④ Less waste - no borated liquid to process

the price

Beginning-of-cycle excess reactivity must be held down by rods and burnable absorbers alone. Rod worth, rod placement and absorber design all get harder, and shutdown margin is harder to prove.

GDC 26 - two independent reactivity control system

## NuScale

- 1 · Control rods
- 2 · boric acid, plus boron pellets that dissolve when the ECCS acts

## i-SMR

- 1 · Control rods
- 2 · none - rods and burnable absorbers must stand in

## BWRX-300

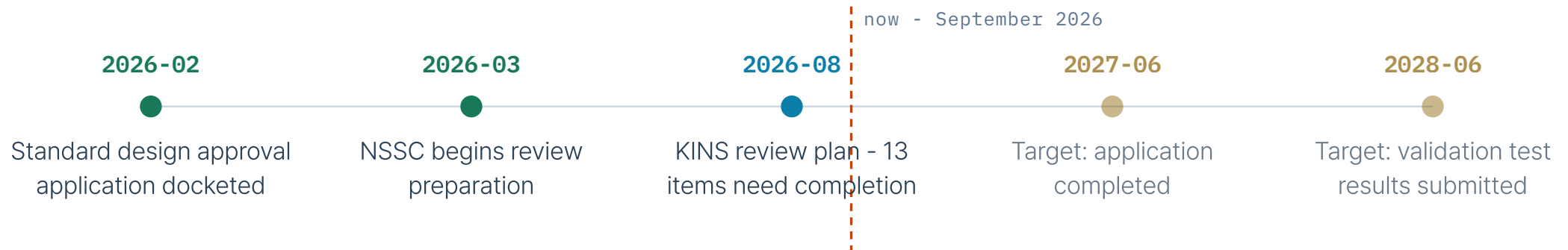
- 1 · Control rods (FMCRD)
- 2 · boric acid the BWR way, plus hydraulic accumulators

**CVCS** Chemical and Volume Control System · **burnable absorber** an absorber that depletes as it burns · **moderator temperature coefficient** how reactivity moves when the moderator heats up · **SDM** shutdown margin · **GDC 26** the criterion requiring two independent reactivity control systems

**Source** This textbook §7.6 · the burnable absorber is  $Gd_2O_3$  mixed into  $UO_2$  rods · shutdown margin  $SDM = \rho_{rod} - \rho_{stuck} - \rho_{defect} - \rho_{unc}$

# i-SMR licensing status, September 2026

- KINS named **the modular layout and the passive safety systems** as focus areas - the parts that large-PWR review experience does not cover
- Watch **the June 2028 milestone for validation test results**. From a laboratory's point of view this is **where the work is**



## What KINS flagged for focus

The modular layout and the passive safety systems - the two things large-PWR review experience does not reach.

## So what is being prepared

Separate-effects tests PASCAL and LAPLACE, integral-effects test ATLAS-PAFS. The core problem is scaling those results up to full size.

Standard design approval - licensing the design itself ahead of a site - KINS Korea Institute of Nuclear Safety - acceptance review checking that an application is complete enough to review • **scaling** the methodology for extrapolating scaled-test results to full size

Source This textbook §7.7, Table 7.4 • as of September 2026 • NSSC and KINS releases and the review plan report are the primary basis

# 08

## Head to head

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The point is not to pick a winner but to find where the same constraint produced different choices. Both chains are internally consistent; they optimised different things.

# Natural versus forced circulation - what sets the flow

- Using a pump in normal operation and cooling passively in an accident are **two different statements**
- i-SMR's **PAFS, PECCS and PCCS are all passive**. The premise is that the safety function holds when the pumps stop

## NuScale

### Natural - flow is bought with height

$$\Delta p = (\rho_c - \rho_h) g H$$

- Flow is fixed by physics; the designer cannot choose it
- Rotating machinery, power, seals and vibration all disappear
- To get the flow you want, lengthen the vessel and cut the loop losses

cooling in an accident

**Passive - DHRS and ECCS**

## i-SMR

### Forced - the pump sets the flow

$$\dot{m} = 2\,658 \text{ kg/s} \cdot \text{RCP} \times 4$$

- Flow becomes a design variable
- The core temperature rise can be held near a large PWR's, 35 K
- Normal operation depends on power - but every safety system is passive

cooling in an accident

**Passive - PAFS, PECCS, PCCS**

**passive** works on physics alone, with no power or operator action · **active** needs power or action · **safety-class equipment** carries a safety function and its own quality requirements

Source This textbook §8.1 · natural:  $\Delta p = (\rho_c - \rho_h) g H$ ; forced: four pumps deliver 2 658 kg/s

# NuScale US460 and i-SMR, spec by spec

- This is why the prompt in chapter 3 said “leave it blank.” Filling a cell with a different generation’s value **to make the table look finished** destroys its credibility

| Item                    | NuScale US460            | i-SMR                         | Note                         |
|-------------------------|--------------------------|-------------------------------|------------------------------|
| Thermal per module      | 250 MWt                  | 540 MWt                       | i-SMR is 2.2×                |
| Electrical per module   | 77 MWe (gross)           | 170 MWe                       |                              |
| Reference plant         | 6 modules · 462 MWe      | 4 modules · 680 MWe           |                              |
| Thermal efficiency      | about 31%                | about 31%                     | both saturated-steam Rankine |
| Primary pressure        | 13.8 MPa                 | 15.5 MPa                      | see §4.5                     |
| Core inlet / outlet     | not confirmed            | 286 / 321 °C                  | US460 not confirmed          |
| Primary flow            | not confirmed            | 2 658 kg/s                    | US460 not confirmed          |
| Circulation             | natural                  | forced · 4 RCPs               |                              |
| Steam generator         | helical once-through ×2  | helical once-through, 8 zones | same type                    |
| Containment design      | 8.3 MPa                  | 5 MPa                         |                              |
| Containment (operating) | vacuum, below 1 psia     | atmospheric class             |                              |
| Fuel and enrichment     | 17×17 · 37 · under 4.95% | 17×17 · 69 · under 5%         | both existing supply chain   |
| Boron · design life     | used · 60 years          | not used · 80 years           |                              |

The two red cells are meant to stay - filling them with US600 values would have completed the table and made it wrong

Source This textbook §8.2, Table 8.1 · NuScale values are US460 · writing the US600-era 264/310 °C into these cells would be wrong

# Safety function by safety function

- Both **make steam, condense it towards a pool and return it by gravity**. Even the valve count matches: **two out, two back**

| Safety function          | NuScale                                     | i-SMR                                       |
|--------------------------|---------------------------------------------|---------------------------------------------|
| Detection                | MPS                                         | RSS - NSSS variables + containment pressure |
| Reactivity control       | rods drop by gravity                        | rods drop by gravity                        |
| Second reactivity means  | boric acid - pellets dissolve when the ECCS | none - burnable absorbers and rods instead  |
| Decay heat removal       | DHRS - 2 trains × 100%                      | PAFS - 36 h cool + 36 h hold                |
| Core cooling (LOCA)      | ECCS - RVV ×2 + RRV ×2                      | PECCS - EDV ×2 + ERV ×2                     |
| What does the condensing | the containment wall itself (submerged)     | the PCCS exchanger → ECT                    |
| Containment pressure     | outer wall submerged, cooled continuously   | PCCS - half the peak within 24 h            |
| Containment isolation    | all passive, fail-closed                    | double isolation valves on the wall         |
| Ultimate heat sink       | pool → 30+ days → air                       | ECT → then air outside the containment      |
| Unattended coping time   | verified to 99 days by analysis             | 72 hours or more                            |

Only the three shaded rows really differ - who condenses, whether a second reactivity means exists, and what ends the coping time

MPS Module Protection System (NuScale) · RSS Reactor Safety System (i-SMR) · **fail-closed** shuts by itself when the signal is lost · ECT emergency cooling tank

Source This textbook §8.3, Table 8.2 · in both designs **nothing after the trip needs power**

# Unattended coping time, and what limits it

- Is it the pool drying out, the batteries running down, or the containment reaching its limit? **The answer differs by design**
- The answer also fixes **how to buy more margin**. If the pool ends it, enlarge the pool. **If containment pressure ends it, a bigger pool changes nothing**

72 h

30 d

99 d

## NuScale



The pool absorbs for 30+ days, then air cooling of the containment wall is claimed indefinitely. The ELAP analysis keeps water above the core to 99 days.

what ends it - pool inventory, then air-cooling capability

## i-SMR



72 hours is an explicit requirement, and PAFS was designed as 36 plus 36. After the ECT is exhausted, natural air circulation outside the containment takes over.

what ends it - the emergency cooling tank inventory

this is not a difference in safety level

It is a difference in how far each vendor drew its design basis. Both meet 72 unattended hours. How much further to demonstrate is a choice - and every extra day costs analysis and testing.

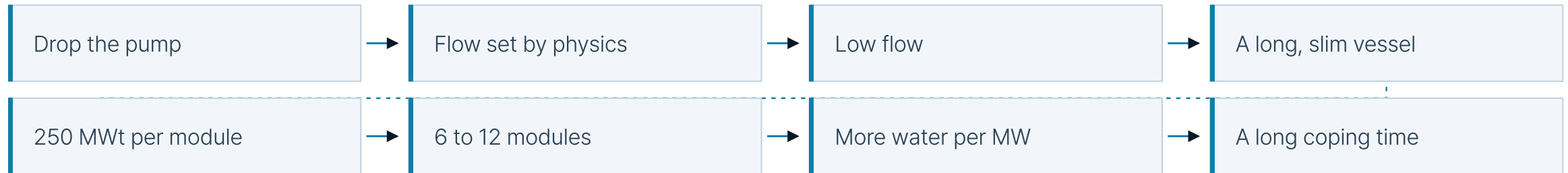
ELAP extended loss of AC power

Source This textbook §8.3.1, Fig. 8.4 · both designs meet 72 unattended hours; how much further to demonstrate is the vendor's choice

# The **chain** of design consequences - two paths

- “**What did this designer optimise?**” If you cannot answer that, you have not read the design
- Find the answer and the rest of the choices **unlock one after another**

**NuScale** It optimised simplicity of the plant



**i-SMR** It optimised reuse of the existing technology base



# 09

## Control case BWRX-300

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A control case against the two PWR designs. Same problems - pipe break, decay heat, containment - solved down the BWR line. Same destination, different route.

# BWRX-300 - overview and main systems

- **300 MWe net, 870 MWt** thermal, **natural** primary circulation, fully passive safety systems, 60-year design life
- A wide remit - **load following (50-100%), hydrogen, district heating, process heat**. Enrichment averages 3.81%, inside the existing supply chain

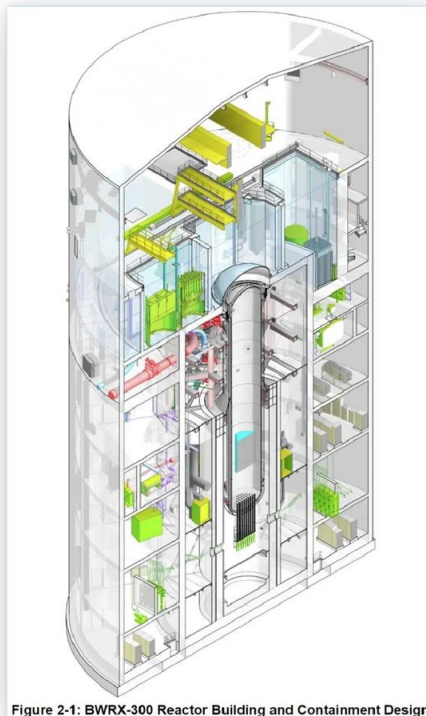


Figure 2-1: BWRX-300 Reactor Building and Containment Design

Reactor building and containment - **mostly below grade**

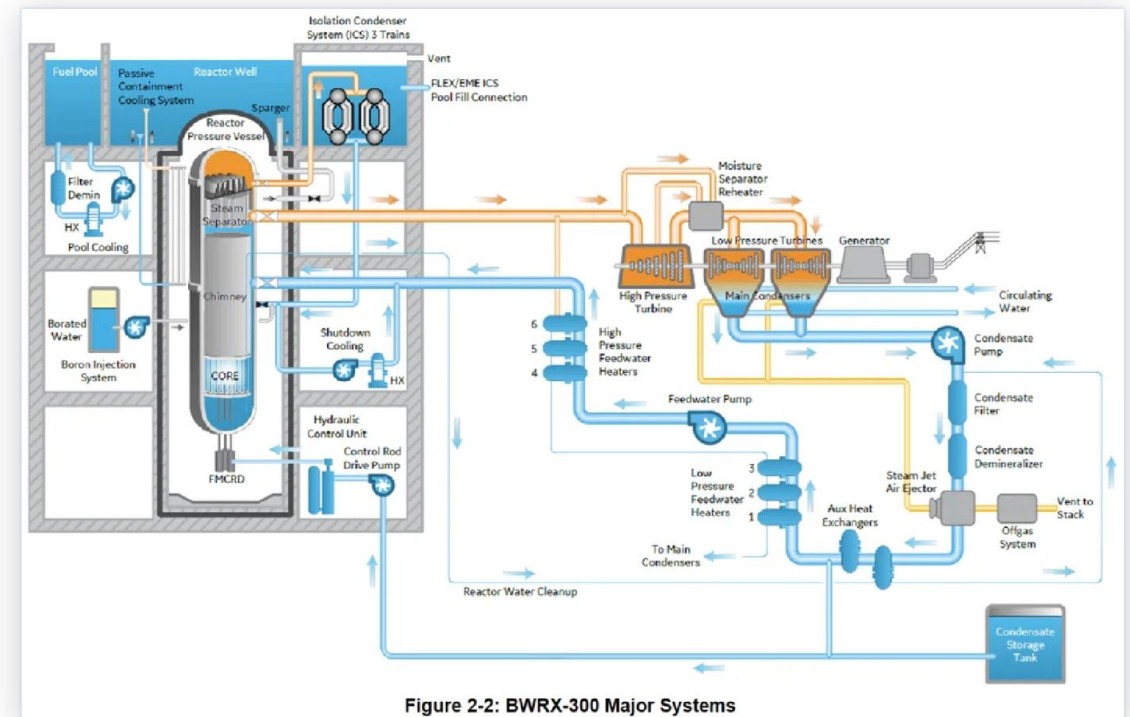


Figure 2-2: BWRX-300 Major Systems

The main systems - reactor vessel, isolation condensers, turbine and condenser

ESBWR the parent design • **isolation condenser** condenses steam inside tubes and returns it • **condenser** turns turbine exhaust back into water

Source This textbook §9.1, Table 9.1 and Figs. 9.1-9.2 • 300 MWe net, 870 MWt

# Chimney-driven circulation and vessel height

- Water carrying voids is light; downcomer water is heavy. **That density difference is the driving head**, and it grows with **the length of the voided run**
- So a **long passage is built above the core on purpose** - the chimney. Hence a vessel 4 m across and 27 m tall

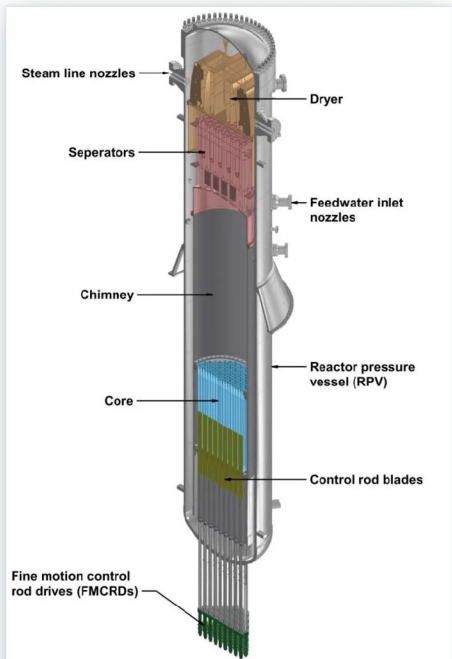
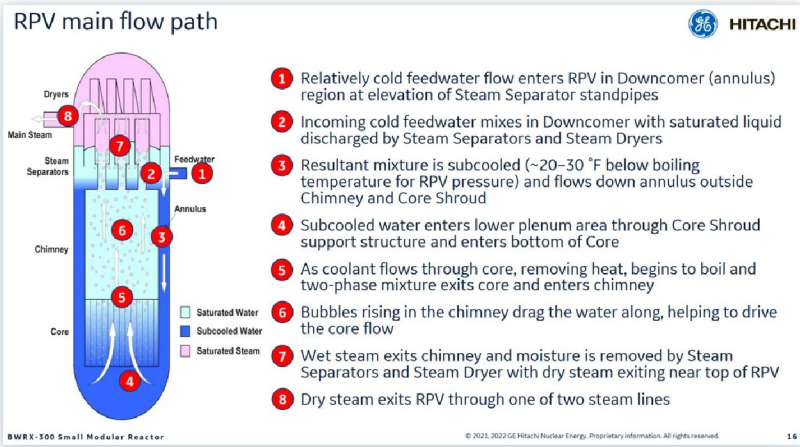


Figure 3-1: BWRX-300 Reactor Pressure Vessel and Internals

Vessel and internals - core, chimney, separator, dryer from the bottom up. **The rods enter from below**



The main flow path - feedwater mixes with separator return in the downcomer, boils in the core, climbs the chimney

$$\Delta p = (\rho_l - \rho_{2\phi}) g H_{ch}$$

|              |               |
|--------------|---------------|
| Thermal      | 870 MWe       |
| Electrical   | 300 MWe (net) |
| Vessel       | 4 m ID · 27 m |
| Core height  | 3.8 m         |
| Wall         | about 136 mm  |
| Enrichment   | 3.81% avg     |
| Control rods | bottom up     |

**chimney** a long passage built above the core on purpose · **two-phase mixture** water carrying steam voids · **shroud** the cylinder separating up-flow from down-flow · **separator and dryer** remove moisture from the steam

Source This textbook §9.2, Figs. 9.3-9.4 · vessel 4 m inside diameter, 27 m tall, 3.8 m core, wall about 136 mm

# Three decisions behind the simplification

- Where an integral SMR erased the problem by removing the pipes, BWRX-300 **keeps them but makes every break isolable** - same destination, different route
- "The device added for safety becomes the cause" is not a rare structure. **Simplifying can point the same way as improving safety**

## RIV

### Reactor isolation valves

Double isolation valves bolted straight to the vessel. Every large penetration carries two, so a break is isolated at once.

Target - no unisolable LOCA for any break above 19 mm

## SRV

### The relief valves were removed

Historically the most common cause of a BWR LOCA was the safety relief valve. The device fitted to hold pressure was the main path for losing coolant.

Instead - a large isolation condenser system and a big steam volume handle overpressure

## Dry

### A dry containment

A metal tank around the vessel and its systems, dry inside - unlike the pressure-suppression containment of earlier BWRs.

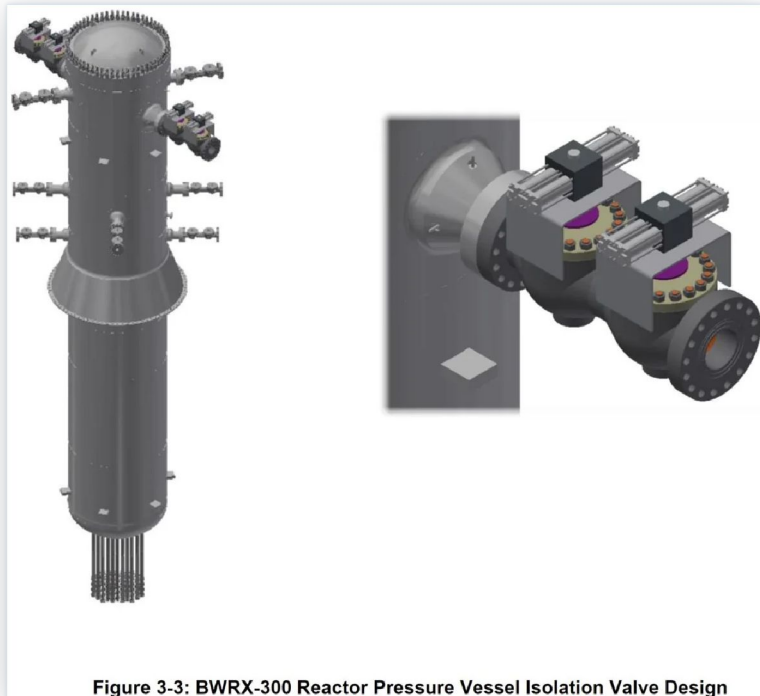
Why it works - thanks to the RIVs. With no unisolable large LOCA, the steam that can arrive is bounded

suppression containment pushes steam into a pool to condense it

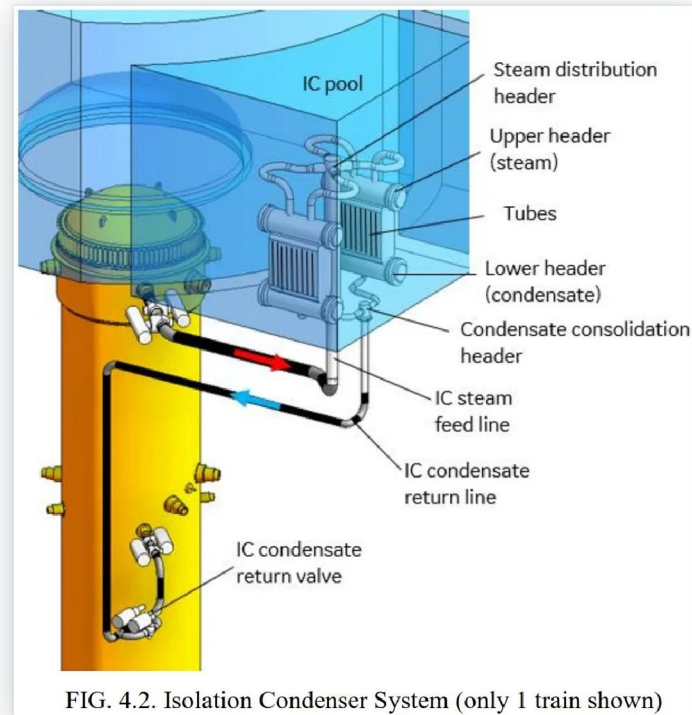
Source This textbook §9.3 · the RIV target is to **eliminate the unisolable LOCA** for any break above DN 19 mm (3/4 in)

# Reactor isolation valves and the dry containment

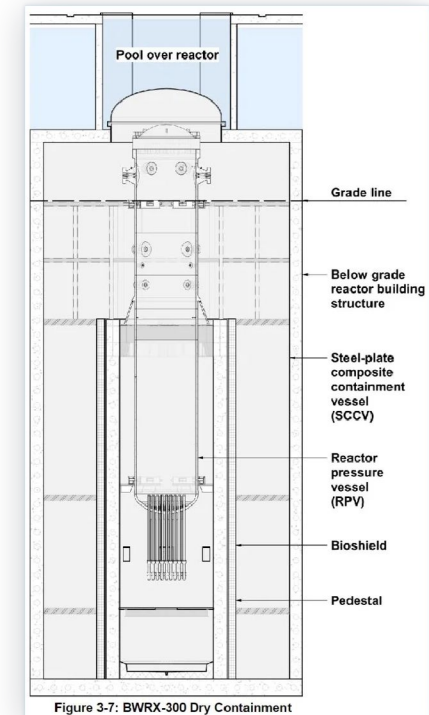
- Every large penetration carries two isolation valves, so a break is shut off at once
- This is what makes the dry containment possible: with no unisolable large LOCA **the steam that can arrive is bounded**, and the containment holds without a suppression pool



The **isolation valve**



**Isolation on the steam line**



The **dry containment**

RIV reactor isolation valve, doubled and bolted to the vessel · **flow restrictor** a throat that caps the escaping flow · **pressure-suppression containment** pushes steam into a pool · **dry containment** holds without that pool

Source This textbook §§9.3.1 and 9.3.3, Figs. 9.5-9.7 · the target is no unisolable LOCA for any break above DN 19 mm (3/4 in)

# Passive decay heat removal - DHRS, PAFS, ICS

- NuScale's DHRS, i-SMR's PAFS and BWRX-300's ICS **work on exactly the same principle**. They differ in **where the steam is taken from**
- The PWR types take it from **the secondary side of a steam generator**; BWRX-300 takes it **straight from the vessel**, because a BWR has no secondary loop

NuScale

## DHRS

steam comes from  
the secondary side of the SG

configuration  
2 trains, each 100%

where it condenses  
condenser submerged in the reactor  
building pool

i-SMR

## PAFS

steam comes from  
the secondary side of the SG

configuration  
36 h cool + 36 h hold

where it condenses  
condenser submerged in the cooling tank

BWRX-300

## ICS

steam comes from  
straight from the reactor vessel

configuration  
3 trains, about 33 MW each

where it condenses  
IC submerged in a pool outside the  
containment

the principle all three share

● Make steam

— ● Condense it in a submerged  
exchanger

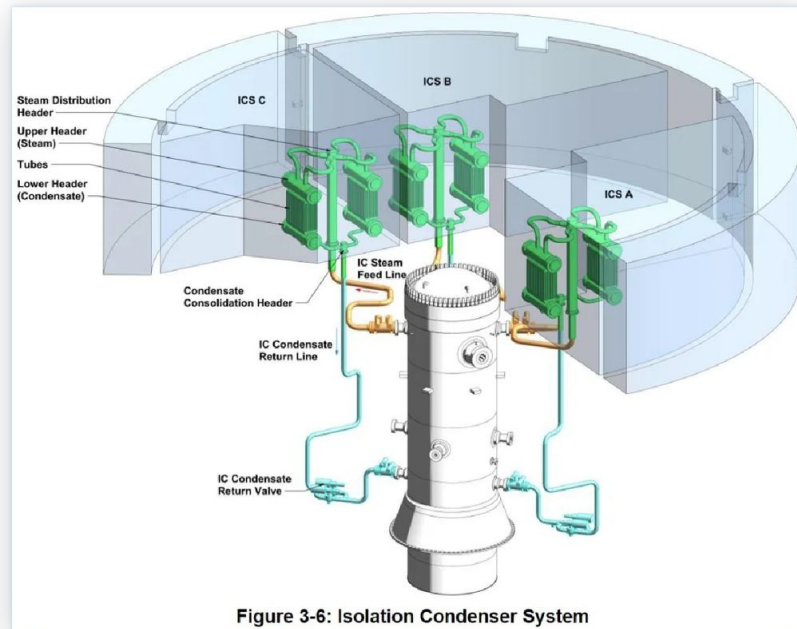
— ● Return it by gravity

— ● A closed loop - nothing added

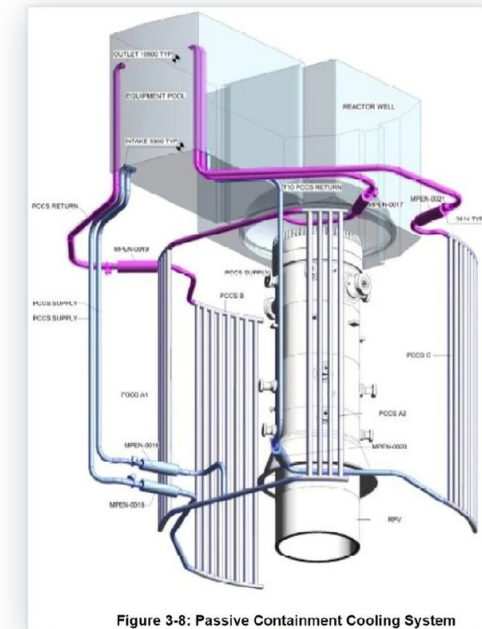
**Source** This textbook §9.4 · three independent trains, about 33 MW each; steam condenses inside the IC tubes and returns to the vessel in a closed loop

# The **isolation condenser** and **passive containment cooling**

- The ICS is the heat sink **when the main condenser is unavailable**. Steam from the vessel condenses inside the IC tubes and returns in a **closed loop**
- The PCCS moves containment heat to the equipment pool by **natural circulation and condensation** in an accident or on loss of active cooling



The isolation condenser system - **three trains** submerged in a pool outside the containment



Passive containment cooling - natural circulation carries the heat to the **equipment pool**

ICS Isolation Condenser System · PCCS Passive Containment Cooling System · **equipment pool** the outside pool that takes the containment heat · **main condenser** the normal-operation steam-to-water exchanger

Source This textbook §§9.4-9.5, Figs. 9.8-9.9 · three independent ICS trains, about 33 MW each

# The fine motion control rod drive

- It solves the BWR problem from chapter 5 - **the rods must be pushed up from below**. A motor turns a ball screw for fine positioning
- A **magnetic coupling** keeps the shaft out of the pressure boundary: no seal, so **one less leak path**

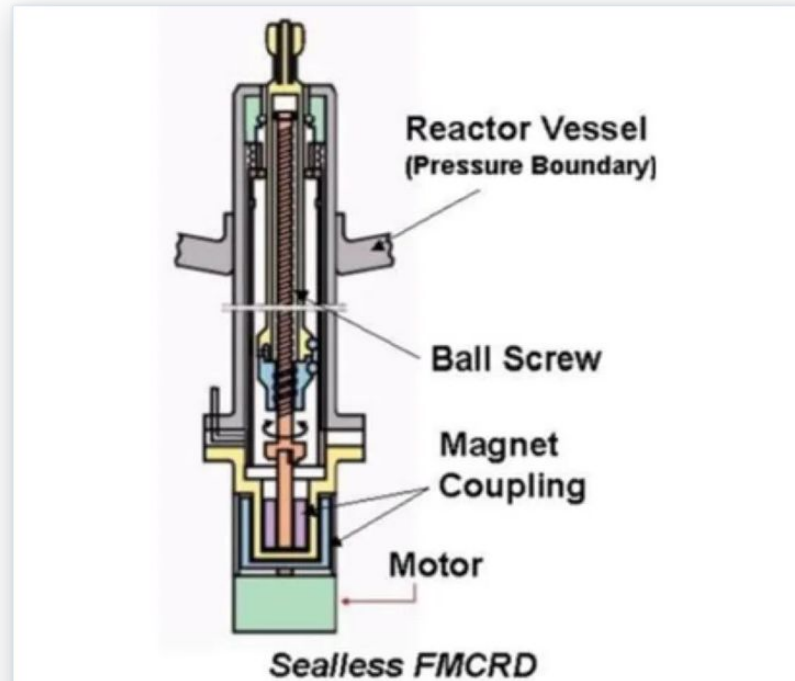


FIG. 2.3. Magnet Coupling Fine Motion Control Rod Drive (FMCRD)

The magnetically coupled FMCRD - a motor turns the ball screw to position the rod

In a PWR the rods **fall by gravity** when power is lost; in a BWR they do not. So the BWR shutdown system guarantees rapid insertion with **hydraulic accumulators**, while the FMCRD handles fine control in normal operation - two parts, two jobs.

An example of GDC 26's "two independent systems" taking a different shape in each line

**FMCRD** Fine Motion Control Rod Drive · **ball screw** turns rotation into linear motion · **magnetic coupling** passes torque across a wall without a shaft through it · **hydraulic accumulator** drives rapid rod insertion with stored gas pressure

**Source** This textbook §9.6, Fig. 9.10 · a magnetic coupling keeps the shaft from **crossing the pressure boundary**

# The three designs, side by side

- What to read is **not the list of answers but the links between them**. Change one choice and three others move

| Item                  | NuScale US460   | i-SMR          | BWRX-300             |
|-----------------------|-----------------|----------------|----------------------|
| Type                  | integral PWR    | integral PWR   | natural-circ BWR     |
| Thermal per module    | 250 MWt         | 540 MWt        | 870 MWt              |
| Electrical per module | 77 MWe          | 170 MWe        | 300 MWe              |
| Primary circulation   | natural         | forced, 4 RCPs | natural, void-driven |
| Operating pressure    | 13.8 MPa        | 15.5 MPa       | about 7 MPa          |
| Steam generator       | helical ×2      | helical ×1     | none                 |
| Large LOCA approach   | pipes removed   | pipes removed  | isolated by valves   |
| Decay heat removal    | DHRS            | PAFS           | ICS - 3 × 33 MW      |
| Containment           | submerged steel | small steel    | dry steel-concrete   |
| Soluble boron         | used            | not used       | used, BWR style      |
| Enrichment            | under 4.95%     | under 5%       | 3.81% avg            |
| Design life           | 60 years        | 80 years       | 60 years             |

Only BWRX-300 has no steam generator - which is why its turbine hall is a radiation area (ch. 5)

**integral PWR** a PWR with the steam generator and pressuriser inside the vessel · **void-driven** circulation by the buoyancy of steam voids made in the core · **isolation** closing valves to shut off a break

**Source** This textbook §9.7, Table 9.2 · NuScale values are US460 · each figure from that design's primary source

# 10

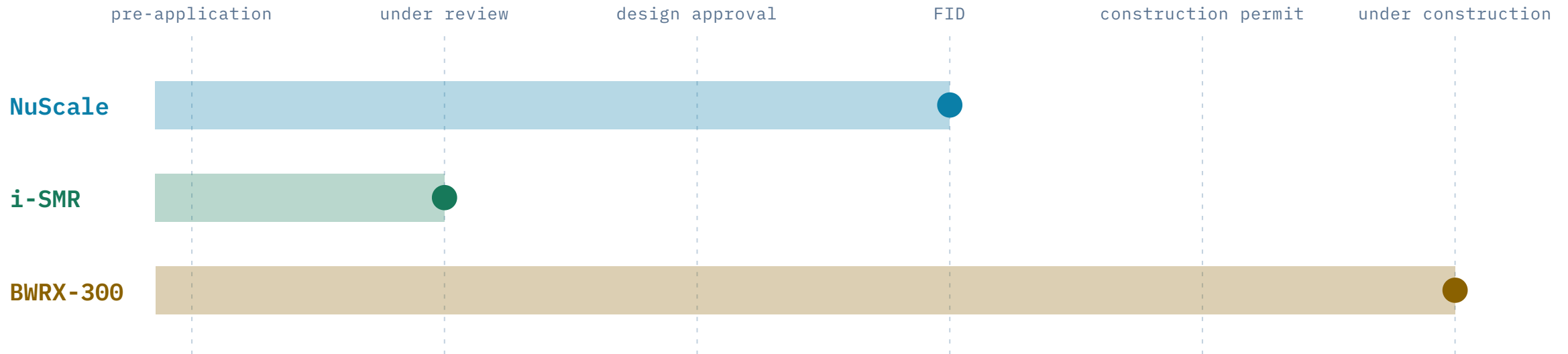
## Wrap-up

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Part I and Part II close together. Finding data and reading a design are the same job. Either way the question is one - where did this value come from, and what must it agree with?

# Licensing path, and where each design stands

- That the three stand in different places is **itself information**. It is why the NEA dashboard scores six dimensions



the marks are where each stood in September 2026

## NuScale

Design approval done (US460, May 2025), and a conditional FID on the first project.

## i-SMR

Applied February 2026, early in review. 13 items to complete; test results due June 2028.

## BWRX-300

Holds a construction licence in Canada and is physically building - furthest along.

**pre-application** aligning issues with the regulator before filing · **design approval** approval of the design itself · **construction permit** permission to build at a named site · **FID** final investment decision

**Source** This textbook §10.1, Fig. 10.1 · **as of September 2026** - this figure ages within six months and is the first thing to revise

# Four shared problems, three answers

- Remove the pipes → the vessel grows → the SG must go inside → it becomes helical → **output is tied to vessel size**
- Remove the boron → the CVCS simplifies → shutdown margin gets harder → **GDC 26's second system needs its own answer**

| The four problems                  | NuScale           | i-SMR                     | BWRX-300                |
|------------------------------------|-------------------|---------------------------|-------------------------|
| What about the large LOCA?         | remove the pipes  | remove the pipes          | keep them, isolate them |
| What pushes the coolant?           | height difference | four pumps                | void buoyancy           |
| Where does the decay heat go?      | pool → air        | cooling tank              | IC pool                 |
| How is reactivity held after trip? | rods + boric acid | rods + burnable absorbers | rods + boric acid       |

|                  |                        |   |                             |   |                             |   |                              |
|------------------|------------------------|---|-----------------------------|---|-----------------------------|---|------------------------------|
| Remove the pipes | the vessel grows       | → | the SG goes inside          | → | it becomes helical          | → | output is tied to the vessel |
| Remove the pump  | flow is set by physics | → | the vessel lengthens        | → | output per module drops     | → | module count rises           |
| Remove the boron | the CVCS simplifies    | → | shutdown margin gets harder | → | GDC 26 needs its own answer |   |                              |

# Four questions to ask of any design document

- **If you cannot see the price, you have not finished reading.** Every design choice has one, and seeing it means you have read the design

## 1 Which source is this number from, and which tier is that source?

Do not cite below the primary tier. Chapters 1-2

## 2 What did this designer optimise?

That answer explains every other choice. Chapter 8

## 3 What is the price of this choice?

If you cannot see it, you are not done reading. Chapters 5-9

## 4 Do these values agree with one another?

One energy balance filters most of it. Chapter 4

# What comes next - non-water designs and system analysis

- Grow to hundreds of nodes and “does the energy close?” is still the question. Only the scale changes

## Non-water-cooled designs

HTGR, SFR, MSR - change the coolant and the thermal-hydraulic problem itself changes. The HALEU bottleneck of §2.4.5 is mostly about these.

- Different properties - density, specific heat, conductivity
- No phase change, or an entirely different one
- Far fewer validated correlations

## System analysis

So far the designs have been read. Next comes computing them in a system code - and the values in Tables 8.1 and 9.2 become the inputs.

- Where did the properties, geometry and boundary conditions come from?
- Traceability is a regulatory requirement
- Do the nodal flows and enthalpy rises add up to the total power?

**system code** a program solving the transient behaviour of a whole reactor system · **node** one computational cell of that system · **HTGR, SFR, MSR** high-temperature gas, sodium fast and molten salt reactors

**Source** This textbook §10.5 · the values in Tables 8.1 and 9.2 become system-code inputs

# Abbreviations (1 of 2) - sources, regulation, projects

| Abbreviation | Expansion                                         | What it means                                            |
|--------------|---------------------------------------------------|----------------------------------------------------------|
| <b>ADAMS</b> | Agencywide Documents Access and Management System | the NRC document system                                  |
| <b>ARIS</b>  | Advanced Reactors Information System              | the IAEA reactor information system                      |
| <b>DSR</b>   | Design Status Report                              | the ARIS design report, filed by the designer            |
| <b>PRIS</b>  | Power Reactor Information System                  | the IAEA operating-reactor database                      |
| <b>NEA</b>   | Nuclear Energy Agency                             | the OECD nuclear agency                                  |
| <b>KINS</b>  | Korea Institute of Nuclear Safety                 | Korea's nuclear safety institute                         |
| <b>RAI</b>   | Request for Additional Information                | the regulator's request for more information             |
| <b>FSER</b>  | Final Safety Evaluation Report                    | the regulator's final safety evaluation                  |
| <b>DCA</b>   | Design Certification Application                  | design certification application, US                     |
| <b>SDA</b>   | Standard Design Approval                          | standard design approval, US and Korea                   |
| <b>FID</b>   | Final Investment Decision                         | final investment decision - not a construction start     |
| <b>EPZ</b>   | Emergency Planning Zone                           | emergency planning zone                                  |
| <b>GDC</b>   | General Design Criteria                           | the US general design criteria                           |
| <b>MCP</b>   | Model Context Protocol                            | the standard for a model to call external tools and data |
| <b>HALEU</b> | High-Assay Low-Enriched Uranium                   | uranium enriched between 5 and 20%                       |

# Abbreviations (2 of 2) - systems and components

| Abbreviation       | Expansion                                            | What it means                                              |
|--------------------|------------------------------------------------------|------------------------------------------------------------|
| <b>NPM</b>         | NuScale Power Module                                 | the NuScale module - containment included                  |
| <b>DHRS</b>        | Decay Heat Removal System                            | decay heat removal, NuScale - 2 trains × 100%              |
| <b>ECCS</b>        | Emergency Core Cooling System                        | emergency core cooling                                     |
| <b>RVV · RRV</b>   | Reactor Vent Valve · Reactor Recirculation Valve     | vent and recirculation valves, NuScale                     |
| <b>IAB</b>         | Inadvertent Actuation Block                          | blocks an ECCS valve from opening by mistake               |
| <b>PAFS</b>        | Passive Auxiliary Feedwater System                   | passive auxiliary feedwater, i-SMR - 36 h + 36 h           |
| <b>PECCS</b>       | Passive Emergency Core Cooling System                | passive emergency core cooling, i-SMR                      |
| <b>PCCS</b>        | Passive Containment Cooling System                   | passive containment cooling                                |
| <b>EDV · ERV</b>   | Emergency Depressurisation · Recirculation Valve     | depressurisation and recirculation valves, i-SMR           |
| <b>RCP · CRDM</b>  | Reactor Coolant Pump · Control Rod Drive Mechanism   | coolant pump and control rod drive                         |
| <b>CVCS</b>        | Chemical and Volume Control System                   | chemical and volume control - much simpler without boron   |
| <b>ICS</b>         | Isolation Condenser System                           | isolation condenser, BWRX-300 - 3 × 33 MW                  |
| <b>RIV · SRV</b>   | Reactor Isolation Valve · Safety Relief Valve        | reactor isolation valve and safety relief valve            |
| <b>FMCRD</b>       | Fine Motion Control Rod Drive                        | fine motion control rod drive, BWRX-300                    |
| <b>ESBWR · PCV</b> | Economic Simplified BWR · Primary Containment Vessel | the parent design of BWRX-300, and the primary containment |

# Source list

- For anything taken from a continuously updated screen, **always record the access date**

| Databases - where you look                                                              | Documents this book cites                                                 | used in |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------|
| <b>IAEA ARIS</b><br>aris.iaea.org<br>design reports, SMR booklet                        | <b>OECD/NEA</b><br>SMR Dashboard 2nd (2024), 3rd (2025-07)                | 2 · 10  |
| <b>US NRC SMR</b><br>nrc.gov/reactors/new-reactors/smr.html<br>US licensing activity    | <b>IAEA ARIS</b><br>i-SMR Design Status Report                            | 7 · 8   |
| <b>NRC ADAMS</b><br>nrc.gov/reading-rm/adams<br>original documents; API exists          | <b>IAEA ARIS</b><br>BWRX-300 Design Status Report (2020)                  | 9       |
| <b>OECD/NEA</b><br>oecd-nea.org – SMR digital dashboard<br>maturity - 79 of 129 tracked | <b>NuScale</b><br>NuScale Power Module Technical Specifications           | 6 · 8   |
| <b>NSSC Korea</b><br>nssc.go.kr<br>primary basis for Korean dates                       | <b>US NRC</b><br>ML25136A329 · press release 25-033 (2025)                | 6       |
|                                                                                         | <b>GE Hitachi</b><br>BWRX-300 General Description                         | 9       |
|                                                                                         | <b>Onweller et al.</b><br>Cited but Not Verified, arXiv:2605.06635 (2026) | 4       |

# "Where did **that number** come from?"

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Answer it **in one line** and the table will hold up in the room.

Value, unit, source, access date, note - five lines is all it takes.

Next comes **the system codes**. The values in Tables 8.1 and 9.2 become **the inputs**.